ProtCOVER: Family-Aware Conformal Prediction for Variant Effect Predictors on Unseen Protein Families

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Abstract. Protein language models and related variant effect predictors (VEPs) are increasingly used to classify missense variants, yet their reliability is typically summarized by benchmark-wide averages, and they are applied to proteins evolutionarily distant from those used for calibration: a family-level distribution shift. Standard split conformal prediction assumes exchangeable variants, and existing VEP calibration schemes condition on residue, gene or domain attributes rather than on homology support. We present ProtCOVER, a conformal procedure that treats homology families, rather than individual variants, as the exchangeable unit. ProtCOVER (i) calibrates at the family level over label-free homology clusters, yielding coverage for a new family under family exchangeability; (ii) adapts to descriptor shift by calibrating conditionally on an estimated family-level density ratio of label-free descriptors (sequence similarity, alignment depth and embedding distance), so that family-weighted coverage holds exactly under family exchangeability, under the estimated shift and under every reweighting in their span, and within a total-variation bound otherwise; and (iii) computes each query's effective calibration support and abstains when it is insufficient. On ProteinGym clinical substitutions with five zero-shot predictors, Mondrian split conformal covers only 0.749–0.790 of pathogenic variants per family at a 0.90 target, even without shift. Under descriptor shift, ProtCOVER's mean worse-class shortfall below the target is 0.001, against 0.009 for group-weighted and 0.032 for hierarchical conformal prediction; on orphan families and held-out embedding regions, which do not shift its descriptors, it matches group weighting (0.001 against 0.004). The ordering holds with deployment-realistic, alignment-free features. ProtCOVER's sets are larger under shift but not at matched coverage. A shift in predictor accuracy that the descriptors cannot see defeats every method, and support-based abstention is no better than random abstention. In a pre-registered evaluation on ClinVar variants added after the benchmark's snapshot, every shift-aware method met the target. Code is publicly available.

Keywords: conformal prediction · variant effect prediction · protein language models · uncertainty quantification · distribution shift · selective prediction · ProteinGym · protein families

1 Introduction

Zero-shot variant effect predictors (VEPs), from protein language models such as ESM-1b [35,7] to alignment-based models such as EVE and GEMME [14,21] and hybrids such as TranceptEVE and PoET [30,39], now rank missense variants with high accuracy on clinical benchmarks [29]. Clinical use, however, needs more than a ranking: variant interpretation under the guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) [34] needs evidence whose error rate is known, and the ClinGen calibration of computational evidence [32] has shown that a single genome-wide calibration can hide large differences between genes [37].

Conformal prediction [41,1] offers distribution-free error control: a prediction set over $\{\text{benign}, \text{pathogenic}\}$ that contains the true label with probability at least $1 - \alpha$. Its guarantee rests on exchangeability between calibration and test data. For VEPs this assumption fails in two ways. First, variants are grouped into proteins and homologous protein families: a predictor's accuracy differs from family to family, and families contribute very different numbers of labeled variants (from 2 to 741 per protein in our data). Pooling variants lets the largest families set the error rate, so the "90%" is a statement about variants from well-studied genes, not about a new protein. Second, the proteins a predictor is deployed on are often evolutionarily distant from, or have shallower alignments than, the proteins in the calibration set: a family-level distribution shift that is visible without labels. We address both as an uncertainty-quantification layer for such predictors: the output is a prediction set with controlled error, not a variant classification or an ACMG/AMP evidence strength.

We make three contributions.

- *Problem and evidence.* We formalize *family-weighted* coverage, the probability that a random variant of a new protein family is covered, and its class-conditional version for pathogenic variants, and show that standard calibration misses it. On ProteinGym clinical substitutions, class-conditional (Mondrian) split conformal, the standard remedy for class imbalance, meets its own variant-level target yet covers only 0.749–0.790 of pathogenic variants per family for four of five predictors at a 0.90 target, with no distribution shift at all; a pre-registered test on ClinVar variants added after the benchmark confirms this (0.873 for ESM-1b). Family-level calibration removes the failure when families are exchangeable.
- *Method and guarantees.* ProtCOVER (i) calibrates at the family level, (ii) calibrates conditionally on a cross-fitted density ratio of label-free family descriptors, and (iii) abstains when calibration support is insufficient. Its main result, Theorem 1, extends the conditional calibration of Gibbs et al. [16] from exchangeable points to exchangeable families: a family-weighted quantile regression whose test family sits at $+\infty$ attains family-weighted coverage exactly under every family-level shift whose density ratio is a nonnegative function in the span of its features, and gives the threshold of hierarchical conformal prediction (HCP) [23] for constant features. With the estimated ratio as a feature, ProtCOVER is exact both under family exchangeability and under the estimated shift, and its loss under any other shift is bounded by the smaller of the bounds for HCP and for reweighting (Corollary 1). Table 1 separates exact guarantees, bounds with unobservable terms, and what is only measured.
- *Evaluation, including where the method does not help.* We test five predictors on five kinds of split (exchangeable; shifted descriptors; orphan families and held-out embedding regions, both defined independently of the descriptors; and a shift in predictor accuracy that violates our assumptions), with deployment-realistic and alignment-free features, four family definitions down to the removal of any detectable homolog, and two pre-registered external evaluations. Under descriptor shift ProtCOVER has the smallest worse-class shortfall below the target (0.001, against 0.009 for group-weighted conformal prediction and 0.032 for HCP); on the two shifts defined independently of the descriptors it matches group weighting. It never exceeds a baseline's shortfall by more than 0.01, and its larger sets under shift are a more conservative operating point, not less efficient sets. A shift that the descriptors cannot see defeats every method, the new variants of the temporal evaluation do not separate the shift-aware methods, and support-based abstention does not beat random abstention.

2 Related Work

Conformal prediction. Split conformal prediction [41,24] gives marginal coverage under exchangeability, and Mondrian versions calibrate within categories [40,10]. Under covariate shift, weighted conformal prediction uses a known or estimated likelihood ratio [38,25], related reweightings handle label shift and general non-exchangeability [33,3], and group-weighted conformal prediction (GWCP) [4] handles shifts in the proportions of known groups; we use GWCP with descriptor bins as a baseline. Localized, robust and multi-source methods reweight by proximity or protect against families of target distributions [18,8,27]. Conditional calibration gives coverage over a class of covariate shifts by quantile regression with the test point imputed [16], and related methods give group-conditional coverage [20,2], all for exchangeable points. For grouped data, Dunn et al. [12] proposed pooling and subsampling for a new group, Lee et al. [23] proved finite-sample guarantees for pooling (HCP), and later work uses a few observations from the test group [28]; Duchi et al. [11] calibrate one coverage threshold per environment, and macro-coverage [5] averages coverage over classes, in the spirit of our family weighting. In protein science, conformal methods have handled the feedback covariate shift of sequence design [13] and mining protein databases for function [6]. Table 1 compares the guarantees: ProtCOVER performs conditional calibration over families with HCP's pooling, and our strongest baseline applies group weighting to the same pooling.

Calibration of variant effect predictors. ClinGen calibrated computational scores to evidence strengths for the computational evidence codes PP3 (supporting pathogenicity) and BP4 (supporting benignity) [32]; subsequent work found that genome-wide calibration masks heterogeneous performance across genes [37] and proposed gene- and domain-aware calibration [9], and RaCoon calibrates ESM-1b at the residue level with conditional score distributions [31]. These methods improve calibration of point probabilities within subgroups seen in training; they do not give distribution-free guarantees, and they condition on gene, domain or residue attributes rather than on homology support relative to the calibration data. Kermut [17] provides

calibrated uncertainty for supervised variant effect regression on deep mutational scans. ProteinGym [29] supplies the benchmark and precomputed scores we use, including those of PLM-based and alignment-based predictors [35,14,21,30,39] and of supervised clinical predictors, which we exclude because they are trained on ClinVar labels [22]. Selective classification [15] motivates our risk-coverage evaluation of abstention.

3 Problem Setup

Families and scores. A protein *family* is a cluster of homologous wild-type sequences obtained without labels (Section 6.1). Family k is the random object $G_k = (D_k, N_k, (X_{k,i}, Y_{k,i})_{i=1}^{N_k})$, where $X_{k,i}$ describes a missense variant, $Y_{k,i} \in \{0, 1\}$ is its label (1 = pathogenic), and $D_k \in \mathcal{D}$ is a *label-free descriptor* of the family: any function of its sequences, alignments, predictor scores, and of the fixed set of training families, but not of labels. A predictor score is turned into $\hat{p}(y \mid x)$ by a base model fitted on separate training families, which we treat as fixed. The nonconformity score is $s(x, y) = 1 - \hat{p}(y \mid x)$ and $S_{k,i} = s(X_{k,i}, Y_{k,i})$. We observe calibration families $G_1, \ldots, G_K$ with labels and a test family G_{K+1} whose descriptor D_{K+1} and variants $X_{K+1,i}$ are observed but whose labels are not.

Coverage targets. The unit of sampling is the family: a new family is drawn from the family-level law (of G_{K+1}), and the question is how often its variants are covered. For a set-valued predictor $C : \mathcal{X} \to 2^{\{0,1\}}$, the per-family coverage of the test family is $\gamma_{K+1}(C) = N_{K+1}^{-1} \sum_i \mathbf{1}\{Y_{K+1,i} \in C(X_{K+1,i})\}$, and our primary target is *family-weighted coverage*

$$\mathrm{Cov}(C) = \mathbb{E}\big[\gamma_{K+1}(C)\big] : \tag{1}$$

draw a new family, then one of its variants uniformly at random; Cov is the probability that this variant is covered. Every family counts once, whatever its number of variants. For example, if one test family has 500 variants and nine others have 5 each, the large family carries over 90% of the weight in variant-weighted coverage, which pools the variants (we report it but do not target it), but a tenth of it in Cov. Its class-conditional version for label y is $\mathrm{Cov}_y(C) = \mathbb{E}\big[N_{K+1,y}^{-1} \sum_{i:Y_{K+1,i}=y} \mathbf{1}\{y \in C(X_{K+1,i})\} \,\big|\, N_{K+1,y} > 0\big]$, where $N_{k,y}$ counts label-y variants of family k; it averages over the families that have label-y variants, not over all families. For a predictor used to flag pathogenic variants, Cov_1 (pathogenic coverage) is the quantity of interest: $1 - \mathrm{Cov}_1$ is the family-weighted *pathogenic miss rate*, the rate at which a pathogenic variant's set excludes the pathogenic label. Coverage for each individual family cannot be guaranteed without assumptions on how families differ [40], so the average over new families is the natural distribution-free target.

Assumptions. We use two alternative assumptions; each result states which one it needs.

A1 (*Family exchangeability.*) $(G_1, \ldots, G_{K+1})$ is exchangeable.

A2 (*Family-level covariate shift.*) $G_1, \ldots, G_K \overset{\text{iid}}{\sim} P$ and, independently, the test family G_{K+1} and any further unlabeled target families are i.i.d. from Q, where $Q \ll P$ and $\frac{dQ}{dP}(G) = w(D)$ for a measurable $w : \mathcal{D} \to [0, \infty)$. Equivalently, the conditional law of a family's variants and labels given its descriptor is the same under P and Q; only the distribution of descriptors shifts.

A1 holds when calibration and test families are drawn by a random split of a pool of families, regardless of dependence between families. It does not require exchangeability of variants within a family, because Cov averages over all variants of the test family (this is weaker than the hierarchical exchangeability of [23], which targets one variant of the new group). A2 formalizes deployment on families that differ from the calibration families in label-free ways, such as being evolutionarily distant or having shallow alignments. It fails when the predictor's accuracy shifts in a way that the descriptors do not see; our stress test S5 constructs such a shift on purpose. Our shifted splits S2–S5 draw test families without replacement from one finite pool, so even where the descriptors capture the shift, A2 holds only approximately and the exact guarantees do not apply; the simulations of Appendix C check the exact case.

4 ProtCOVER

ProtCOVER has four components: (i) family-level calibration; (ii) adaptation to descriptor shift; (iii) a support gate; and (iv) class-conditional versions of each. Its default (ii) is conditional calibration with features $(1, \hat{w})$, for an estimated density ratio $\hat{w}$. Two variants replace it: *ProtCOVER-W* reweights the families by $\hat{w}$, and *ProtCOVER-D* conditions on the descriptors instead of $\hat{w}$.

(i) Family-level calibration. Let $\hat{F}_k = N_k^{-1} \sum_i \delta_{S_{k,i}}$ be the empirical distribution of calibration scores of family k. Given nonnegative family weights $\omega_1, \ldots, \omega_{K+1}$, set $p_k = \omega_k / \sum_{j=1}^{K+1} \omega_j$ and

$$\hat{\mu} = \sum_{k=1}^{K} p_k \hat{F}_k + p_{K+1}\, \delta_{+\infty}, \qquad \hat{q} = \mathrm{Quant}_{1-\alpha}(\hat{\mu}), \qquad C(x) = \{y \in \{0,1\} : s(x,y) \le \hat{q}\}, \tag{2}$$

where $\mathrm{Quant}_\tau(\mu) = \inf\{u : \mu((-\infty, u]) \ge \tau\}$. With $\omega \equiv 1$ every family, calibration or test, has the same total weight, spread evenly over its variants. The test family's weight sits at $+\infty$ because its labels, and hence its scores, are unknown: the guarantee holds for the threshold computed with its true scores, and moving them to $+\infty$ can only raise the threshold. This is the hierarchical conformal prediction (HCP) threshold of [23], which equals the CDF pooling of [12] up to the level adjustment. Variant-level split conformal instead gives every calibration variant weight one, so families with hundreds of variants set the quantile.

(ii) Adapting to descriptor shift: conditional calibration on an estimated density ratio. Equation (2) uses one threshold for every test family. ProtCOVER lets the threshold depend on the test family's label-free descriptor through a feature map $\Phi : \mathcal{D} \to \mathbb{R}^p$ with $\Phi_1 \equiv 1$, spanning $\mathcal{F} = \{\Phi(\cdot)^\top b : b \in \mathbb{R}^p\}$. With $\tau = 1 - \alpha$ and the pinball loss $\rho_\tau(u) = \max\{\tau u, (\tau - 1)u\}$, let

$$L(b) = \sum_{k=1}^{K} \frac{1}{N_k} \sum_{i=1}^{N_k} \rho_\tau\big(S_{k,i} - \Phi(D_k)^\top b\big) - \tau\, \Phi(D_{K+1})^\top b, \qquad \hat{q} = \sup\{\Phi(D_{K+1})^\top b :\ b \in \arg\min L\}, \tag{3}$$

with $\hat{q} = +\infty$ if L is unbounded below or if the fitted value $\Phi(D_{K+1})^\top b$ is unbounded over $\arg\min L$, and $C(x) = \{y : s(x,y) \le \hat{q}\}$. This is a family-weighted quantile regression in which, as in (2), each calibration family has total weight one and the test family's scores sit at $+\infty$ (the last term); with $\Phi \equiv 1$ it is the HCP threshold, except that when the pooled distribution $\hat{\mu}$ reaches $1 - \alpha$ exactly at that threshold it takes the next pooled calibration score, or $+\infty$ if there is none. L is bounded below iff $-\tau\Phi(D_{K+1}) = \sum_j a_j \Phi_j$ for some weights $(\tau - 1)/N_{k(j)} \le a_j \le \tau/N_{k(j)}$ on the calibration scores j; otherwise the query's features cannot be balanced by the calibration families, for example when its estimated ratio far exceeds theirs, and the set is $\{0,1\}$: a built-in abstention. The default feature map is $\Phi = (1, \hat{w})$, where $\hat{w}$ is an estimate of the density ratio $w = dQ/dP$ of descriptors, so the threshold can rise for families that are rare among the calibration families and frequent among the target families. Of the two variants, *ProtCOVER-W* instead reweights the families in (2) with $\omega_k = \hat{w}(D_k)$, $k \le K + 1$ (computable because D_{K+1} is label-free), and *ProtCOVER-D* uses the standardized support descriptors themselves as features and needs no target families.

The ratio comes from a logistic classifier separating calibration families (class 0) from target families (class 1) on standardized descriptors: $\hat{w}(d) = \frac{n_0}{n_1} \exp\{\hat{f}(d)\}$, with $\hat{f}$ the fitted log-odds and n_0, n_1 the class sizes, L_2 penalty chosen by cross-validated log loss on this membership task (no variant labels), rescaled to mean one over the fitting calibration families and clipped to $[1/c, c]$ ($c = 20$). The descriptors measure sequence similarity (maximum identity of the family to any training protein over significant MMseqs2 [36] hits), alignment depth ($\log N_{\mathrm{eff}}$ and N_{eff} per position, with N_{eff} the effective number of sequences in the alignment) and embedding distance (cosine distance of mean-pooled ESM2 [26] embeddings to the nearest training protein), together with log length and log number of scored variants. *Cross-fitting:* the calibration families are split once at random into a third that fits $\hat{w}$ and two thirds used for calibration, and the target families into five folds; the ratio for queries in fold j is fitted against the target families outside fold j, so each fold has its own ratio and its own guarantee, conditional on that ratio (Lemma 1).

Computation. We solve (3) through its dual linear program, which has one bounded variable per calibration score but only p equality constraints; test families change only the right-hand side, so HiGHS [19] re-solves each from the previous basis (a few milliseconds). To obtain the supremum in (3) when $\arg\min L$ is not a single point, the test term's weight is raised from τ to $\tau + 10^{-4}$: every solution of the perturbed program has a fitted value at least the supremum, and an infeasible perturbed dual (an unbounded perturbed program) gives $+\infty$ (Appendix A). The computed sets therefore contain those of (3), and keep every guarantee by Lemma 2; an independent primal computation of 1,800 thresholds under tighter solver tolerances changes no audited set (Appendix B).

(iii) Support gate. The gate returns $\{0, 1\}$ for every variant of a query family (it abstains) when few calibration families resemble it, measured by a kernel effective sample size (ESS) of the support descriptors below m, or when the ratio puts more than $\pi_{\max}$ of ProtCOVER-W's quantile mass on the query itself; its parameters are tuned on an inner split of the training families (Appendix B).

(iv) Class-conditional versions. Every component has a Mondrian version: for each candidate label y, the threshold $\hat{q}_y$ is computed from label-y calibration variants only (weights $1/N_{k,y}$; families with $N_{k,y} = 0$ dropped), and $y \in C(x)$ iff $s(x, y) \le \hat{q}_y$. These target Cov_0 and Cov_1 separately and are the versions we compare throughout.

5 Guarantees

All results are finite-sample and hold for any fixed score function, so they do not depend on the quality of the base model or of the predictor; proofs are in Appendix A. Each concerns the expected coverage of a random variant of a new family, not the coverage of every family. Table 1 lists what each one does and does not cover.

Proposition 1 (Family-level calibration). *Under A1, the set* (2) *with* $\omega \equiv 1$ *satisfies* $\mathrm{Cov}(C) \ge 1 - \alpha$.

Proposition 1 is Theorem 1 of [23] with the test point replaced by a uniformly random variant of the test family, which removes the need for exchangeability within families.

Theorem 1 (Coverage over a class of family-level shifts). *Let P be a law of families and Φ fixed. For $f \in \mathcal{F}$ with $f(D) \ge 0$ P-almost surely and $0 < \mathbb{E}_P f(D) < \infty$, let Q_f be the family law with $dQ_f/dP = f(D)/\mathbb{E}_P f(D)$. If $G_1, \ldots, G_K$ are i.i.d. P and, independently, $G_{K+1} \sim Q_f$, the set* (3) *satisfies* $\mathrm{Cov}(C) \ge 1-\alpha$. *In particular ($f \equiv 1$),* $\mathrm{Cov}(C) \ge 1 - \alpha$ *under A1.*

Corollary 1 (Any family-level shift). *For a test family with any law Q,* $\mathrm{Cov}(C) \ge 1-\alpha-\inf_f d_{\mathrm{TV}}(Q, Q_f)$, *the infimum over the f of Theorem 1. If $\Phi = (1, \hat{w})$ for a ratio estimate $\hat{w}$ independent of the families, with $0 < \mathbb{E}_P \hat{w} < \infty$ and $d\hat{Q}/dP = \hat{w}(D)/\mathbb{E}_P \hat{w}(D)$, then*

$$\mathbb{E}\big[\gamma_{K+1}(C) \,\big|\, \hat{w}\big] \;\ge\; 1 - \alpha - \min\big\{d_{\mathrm{TV}}(Q, P),\; d_{\mathrm{TV}}(Q, \hat{Q})\big\}.$$

Here $d_{\mathrm{TV}}(Q, P)$ bounds the loss of family-level calibration alone (the same argument applied to Proposition 1) and $d_{\mathrm{TV}}(Q, \hat{Q})$ that of reweighting by $\hat{w}$ (Proposition 2), so conditioning on the estimated ratio is never worse, in these bounds, than either. Both distances are unobservable, so the corollary explains the method's behavior but does not certify coverage under an arbitrary shift. Conditioning on the ratio is exact when families are exchangeable, exact when $\hat{w}$ is the true ratio up to scale, and exact for every shift whose ratio is a nonnegative function in the span of 1 and $\hat{w}$. With descriptor features (ProtCOVER-D), it is exact for every shift whose density ratio is a nonnegative affine function of the descriptors, and it needs no target families at all.

What is new. Table 1 places these results among related guarantees. Theorem 1 takes two technical steps beyond the conditional calibration of Gibbs et al. [16]. First, it moves from exchangeable points to exchangeable families by pooling: each calibration family enters the quantile regression as its whole empirical score distribution with total weight one, and the test family enters at $+\infty$, so that the program's first-order condition is a statement about family-weighted coverage and constant features give HCP's threshold. Applying Gibbs et al. to one random variant per family, the analogue of subsampling once [12], already gives family-weighted coverage $1 - \alpha$ over $\mathcal{F}$, because the subsampled pairs are exchangeable points; we have not seen this combination stated and use it as a baseline. It discards all but one calibration variant of each family, and the validity of the pooled program does not follow from it, as that of HCP does not follow from subsampling [23]. Second, the threshold is the largest optimal fitted value. This adapts Gibbs et al.'s Proposition 6, which imputes an upper bound for the unknown scores and assumes min-norm solutions: we let the bound go to $+\infty$, need no particular solution, and treat unbounded programs, which give the set $\{0, 1\}$. Conditioning on the span of a constant and a cross-fitted ratio (Gibbs et al. note a single estimated shift as a one-element class) then gives Corollary 1. Duchi et al. [11, Cor. 7.27] regress one coverage threshold per environment on environment features and bound the probability that a new environment's coverage falls below $1 - \alpha$; this implies family-level coverage of at least $(1 - \alpha)(1 - \delta)$ under the corresponding shifts, a conservative level, and does not pool the calibration variants.

Table 1. Guarantees of related methods and of ProtCOVER's versions. Every listed guarantee is a finite-sample lower bound of $1-\alpha$ on the stated target under the stated conditions; "bound" means a lower bound reduced by a total-variation distance that cannot be estimated from unlabeled data. ProtCOVER's versions hold for Cov and, class-conditionally, for Cov_y given $N_{K+1,y} > 0$ (Proposition 3); with cross-fitted ratios they hold conditionally on each fold's ratio (Lemma 1).

Method	Unit	Target	Exact under	Target data
Split, Mondrian [24,40]	variant	a new variant (of class y)	exchangeable variants	none
Weighted [38,25]	variant	a new variant	covariate shift with the true ratio; estimated: bound	unlabeled
GWCP [4]	variant	a new variant	shift of the proportions of known groups	group proportions
Conditional calibration [16]	variant	a new variant	every shift with ratio $f(X)$, $f \in \mathcal{F}$, $f \geq 0$	none
HCP, subsampling [23,12]	family	a variant of a new family	A1	none
Multi-environment [11, Cor. 7.27]	environment	a new environment, w.p. $1-\delta$	shifts in a class of environment features	none
Conditional calibration, one variant per family	family	Cov	every shift with ratio $f(D)$, $f \in \mathcal{F}$, $f \geq 0$ (incl. A1)	as $\mathcal{F}$ needs
ProtCOVER-W (Prop. 2)	family	Cov, Cov_y	A2 with the true ratio; estimated: bound	unlabeled families
ProtCOVER-D (Thm. 1)	family	Cov, Cov_y	every shift with ratio affine in the descriptors, ≥ 0 (incl. A1)	none
ProtCOVER (Thm. 1, Cor. 1)	family	Cov, Cov_y	every shift with ratio $a + b\,\hat{w}(D) \geq 0$ (incl. A1); otherwise bound	unlabeled families
Gate (Lemma 2)	family		never lowers coverage; nothing on the retained families	
Not guaranteed	coverage of any given family; the splits S2–S5, where A2 holds at best approximately and is violated in S5			

Proposition 2 (Reweighting variant). *Under A2, the set* (2) *with* $\omega_k = w(D_k)$, $k \leq K+1$, *satisfies* $\mathrm{Cov}(C) \geq 1-\alpha$. *With* $\omega_k = \hat{w}(D_k)$ *for* $\hat{w}$ *independent of the families,* $\mathbb{E}[\gamma_{K+1}(C) \mid \hat{w}] \geq 1-\alpha-d_{\mathrm{TV}}(Q,\hat{Q})$.

Proposition 2 lifts weighted exchangeability [38] and its estimated-weight bound [25, Thm. 3] from variants to families.

Lemma 1 (Cross-fitting). *For each fold* j, *the ratio* $\hat{w}_j$ *is a function only of the fitting calibration families and of the target families outside fold* j, *rescaling and clipping included. Under A2 it is therefore independent of the calibration families used in* (2) *and* (3) *and of every family in fold* j, *and Corollary 1 and Proposition 2 hold for the queries of fold* j *conditionally on* $\hat{w}_j$. *If instead all non-training families are exchangeable, as under a random split, then conditionally on the families that fit* $\hat{w}_j$ *the remaining calibration families and each family of fold* j *are exchangeable, and Theorem 1 with* $f \equiv 1$ *holds conditionally on* $\hat{w}_j$.

Proposition 3 (Class-conditional versions). *For* $y \in \{0,1\}$ *with* $\mathbb{P}(N_{K+1,y} > 0) > 0$, *the Mondrian versions of Propositions 1 and 2, Theorem 1, Corollary 1 and Lemma 1 hold with* Cov_y *in place of* Cov *and with* Q, P, Q_f, $\hat{Q}$ *conditioned on* $N_{K+1,y} > 0$.

Lemma 2 (Enlargement). *If* $C'(x) \supseteq C(x)$ *for all* x *almost surely, where* C' *may depend on all observed data, then* $\mathrm{Cov}(C') \geq \mathrm{Cov}(C)$ *and* $\mathrm{Cov}_y(C') \geq \mathrm{Cov}_y(C)$.

The gate (iii) only replaces a set by $\{0,1\}$, so by Lemma 2 it preserves every guarantee above, whatever its parameters; it gives no guarantee for the families it retains.

6 Experiments

6.1 Data, predictors, and families

We use the ProteinGym v1.3 clinical substitution benchmark [29]: 62,727 ClinVar missense variants in 2,525 human proteins, 51.0% pathogenic, with a median of 12 labeled variants per protein (maximum 741). We take ProteinGym's precomputed zero-shot scores for five unsupervised predictors: ESM-1b (protein language model),

EVE and GEMME (alignment-based), TranceptEVE (hybrid), and PoET (retrieval-augmented language model), oriented so that higher means more pathogenic; our per-protein mean AUCs match ProteinGym's published values (0.892–0.920) to three decimals. Supervised clinical predictors are excluded because they are trained on ClinVar.

Families are MMseqs2 [36] clusters of wild-type sequences at 30% identity and 80% bidirectional coverage: 2,081 families, 1,805 of them single proteins. Three alternative definitions serve as sensitivity checks (Appendix B): 50% identity (2,275 families), single-linkage components of significant homology (1,939), and, strictest, components of *any* significant hit (1,024, the largest with 867 proteins), so that no test protein has a detectable homolog, even through a shared domain, among the training or calibration proteins. Single linkage through shared domains chains unrelated proteins together, so this last definition is a stress test for leakage, not a biological family. Descriptors are computed without labels (Section 4); a family is an *orphan* if no member has a significant hit at $\geq 30\%$ identity outside the family (1,276 families).

6.2 Protocol

Each of ten seeds splits families into 40% training, 30% calibration and 30% test (832, 624 and 625 families). **S1** draws all three at random. **S2** keeps the S1 training families and samples test families with probability proportional to $\exp(-\beta z)$, z the standardized mean of maximum identity to training and $\log N_{\text{eff}}$, $\beta \in \{0.5, 1, 2\}$; it shifts the very descriptors that ProtCOVER and GWCP use, so it favors them by construction. Two shifts are defined independently of those descriptors: **S3** draws test families from orphan families (mean maximum identity to training 0.048 against 0.236 for calibration families), and **S4** holds out whole regions of protein space, k-means clusters ($k = 20$) of the leading principal components of family-mean ESM2 embeddings, taken in random order until the test set is full; the last region is split at random, so a strict variant, **S4s**, skips every region that would overfill the test set and draws training and calibration families from the rest (Appendix B). **S5** is a stress test that violates A2 on purpose: it keeps the S1 training families and samples test families with weight $\exp(-z)$, where z is a family's standardized within-family AUC averaged over predictors, so test families are those on which predictors err more, a shift that no label-free descriptor encodes. S4, S4s and S5 were declared in the configuration before they were first run.

The base model, fitted on training families only, is a logistic regression on the raw score (clipped to the training 0.1% and 99.9% quantiles) and on the score standardized within its protein over the protein's scored variants; the latter is label-free but needs the protein's whole variant panel, so a deployment-realistic run uses the raw score only and drops the variant-count descriptor. Its test AUC is 0.873–0.929. Baselines are variant-level split conformal (B1) and its class-conditional (Mondrian) version (B2); HCP pooling and Dunn et al.'s single and repeated subsampling (B3) [23,12]; GWCP (B4) [4] with nine bins (maximum identity zero, low or high, crossed with $\log N_{\text{eff}}$ terciles) and target proportions from the test families; and support-only and random abstention at the gate's abstention rate (B5). ProtCOVER uses 416 of the calibration families in (2)–(3) and the rest to fit $\hat{w}$; ProtCOVER-D uses all calibration families and the support descriptors. The gate is tuned on a shifted inner split of the training families under a pre-declared 10% abstention budget. Test families are used only for evaluation, and only their label-free descriptors enter $\hat{w}$. Appendix B lists every setting.

Comparison criterion. Because family-level calibration can err in either class (for some predictors it over-covers pathogenic and under-covers benign variants), we compare class-conditional methods by the *worse-class shortfall* $\max\{0,\, 1 - \alpha - \min(\text{Cov}_0, \text{Cov}_1)\}$, per predictor and setting, and treat differences below 0.01 as no difference, the criterion that we pre-registered for the external evaluation. The margin is a statistical, not a clinical, threshold: it lies between the median half-width of a cell's bootstrap interval for class-wise coverage (0.005) and the median seed-to-seed standard deviation (0.015). Intervals are 95% family-bootstrap intervals that resample test families within each seed (paired across methods); they are conditional on the splits, and since all seeds resplit one pool of families, inference is limited to this benchmark. Unless stated otherwise $\alpha = 0.1$.

External evaluations. Two evaluations calibrate on benchmark variants and test on ClinVar variants that played no part in development. For each, the data filters, endpoints and hypotheses were pre-registered in the public repository before any method was run on them; both use the ESM-1b scores of Brandes et al. [7] and, as primary analysis, the deployment-realistic base model (Appendix B). The *temporal* evaluation tests on 14,510 variants in 1,731 benchmark proteins that entered ClinVar after the benchmark's snapshot,

Table 2. Worse-class shortfall below the 0.90 target, $\max\{0, 0.9 - \min(\mathrm{Cov}_0, \mathrm{Cov}_1)\}$, of class-conditional family-level methods at $\alpha = 0.1$: mean (maximum) over the predictor-setting cells of each row. Lower is better; 0.000 means the target is met in every cell. The last two rows give mean set sizes over the benchmark's shifted cells, at $\alpha = 0.1$ and interpolated to pathogenic coverage 0.90.

Evaluation	HCP	GWCP	ProtCOVER-W	ProtCOVER-D	**ProtCOVER**
Benchmark, five predictors (development data)					
S2, descriptor shift, $\beta \in \{0.5, 1, 2\}$	0.032 (0.102)	0.009 (0.027)	0.002 (0.009)	0.002 (0.010)	0.001 (0.005)
S3, orphan families	0.013 (0.024)	0.004 (0.011)	0.001 (0.006)	0.001 (0.006)	0.002 (0.010)
S4, held-out embedding regions	0.006 (0.013)	0.004 (0.005)	0.008 (0.014)	0.005 (0.015)	0.000 (0.000)
S4s, whole embedding regions only	0.008 (0.015)	0.006 (0.011)	0.004 (0.008)	0.000 (0.000)	0.000 (0.001)
S5, label-based shift (violates A2)	0.110 (0.122)	0.104 (0.115)	0.107 (0.113)	0.099 (0.111)	0.102 (0.109)
Robustness checks, S2–S4 pooled					
Raw score only, no variant count	0.019 (0.080)	0.007 (0.024)	0.003 (0.013)	0.004 (0.016)	0.001 (0.008)
Raw score, alignment-free descriptors	0.019 (0.080)	0.019 (0.080)	0.015 (0.068)	0.016 (0.070)	0.011 (0.060)
Families: MMseqs2 50% identity	0.023 (0.112)	0.009 (0.030)	0.004 (0.027)	0.002 (0.012)	0.002 (0.015)
Families: homology components	0.020 (0.096)	0.009 (0.036)	0.004 (0.014)	0.001 (0.012)	0.002 (0.015)
Families: any significant hit	0.024 (0.130)	0.009 (0.047)	0.002 (0.009)	0.001 (0.006)	0.000 (0.004)
External ClinVar variants after the snapshot, ESM-1b, S2–S4 (pre-registered)					
Raw score only (primary)	0.004 (0.020)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Main base model (secondary)	0.005 (0.023)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Mean set size, benchmark S2–S4					
At $\alpha = 0.1$	1.296	1.310	1.335	1.324	1.357
At matched $\mathrm{Cov}_1 = 0.90$ (24 cells)	1.334	1.327	1.333	1.317	1.330

in the benchmark's splits: new variants and labels, not new proteins. Its hypotheses are that Mondrian split under-covers pathogenic variants per family in S1 (H1), that HCP covers both classes (H2), that ProtCOVER-D's shortfall over the shifted settings is at most HCP's and GWCP's (H3), and that abstention does not reduce the error of retained singleton sets (H4). ProtCOVER was added to H3 after the development results had singled it out, though before the run, so for ProtCOVER H3 is an added comparison rather than an original hypothesis.

6.3 Results

Variant-level calibration misses pathogenic variants per family. In S1 (Appendix Fig. 1), variant-level split conformal has variant-weighted coverage 0.894–0.901, close to its target, but family-weighted coverage 0.880–0.885. The gap is concentrated in the pathogenic class: Mondrian split conformal (B2), the standard remedy for class imbalance, attains variant-weighted pathogenic coverage 0.893–0.905 for four predictors yet covers only 0.749–0.790 of pathogenic variants per family. The 5% of test families with the most pathogenic variants hold 50% of them, and their pathogenic variants are easier: under B2, families with fewer than five pathogenic variants have pathogenic coverage 0.660–0.727 against 0.899–0.921 for families with at least twenty. TranceptEVE is the exception (0.928), with the error moved to the benign class. Family-level class-conditional calibration attains 0.898–0.904 (pathogenic) and 0.898–0.903 (benign), as Proposition 1 guarantees, and ProtCOVER 0.901–0.906 and 0.900–0.907: without shift, conditioning on the ratio costs nothing.

Under descriptor shift ProtCOVER has the smallest shortfall; on the other shifts it matches group weighting. Table 2 is the main comparison. We read S2, which shifts the descriptors that ProtCOVER and GWCP use, separately from S3 and S4, which do not. In S2, the mean worse-class shortfall is 0.032 for HCP, 0.009 for GWCP, 0.002 for ProtCOVER-W, 0.002 for ProtCOVER-D and 0.001 for ProtCOVER, whose shortfall is smaller than GWCP's by more than the 0.01 margin in 5 of the 15 S2 cells and than HCP's in 14. On S3 and S4 it is 0.001, against 0.004 for GWCP and 0.009 for HCP; ProtCOVER's worse-class coverage exceeds GWCP's with a paired interval above zero in 8 of 10 cells, but only by -0.001 to $+0.016$, so within the margin it matches GWCP there. Over all 25 shifted cells, ProtCOVER's worst cell is 0.8898 (HCP 0.798, GWCP 0.873), and its shortfall exceeds no baseline's by more than the margin in any cell. Conditioning, rather than reweighting, is what keeps this safe: on S4, where family-level calibration alone is close to the target and the estimated weights mainly add variance, ProtCOVER-W falls short by 0.008 on average (GWCP 0.004), while ProtCOVER, which keeps the constant feature and so cannot do worse than HCP in the bound of Corollary 1, meets the target in every S4 cell. The ordering holds at $\alpha = 0.05$ and 0.2 (mean shortfall 0.000 and 0.003 for ProtCOVER, 0.006 and 0.012 for GWCP; paired class-wise comparisons in Appendix Tables 6–7).

The price is a more conservative operating point, not less efficient sets. ProtCOVER adapts how conservative it is to the shift: its mean worse-class coverage is 0.902 in S1 (HCP 0.900) but 0.906 over the shifted cells (GWCP 0.893, HCP 0.877; Appendix Fig. 2). Its sets under shift are therefore larger (mean size 1.36, GWCP 1.31, HCP 1.30), but at matched pathogenic coverage the family-level methods need about the same set size (1.33, 1.33 and 1.33 for ProtCOVER, GWCP and HCP over the 24 cells where all reach it; Appendix Fig. 3 and Table 9). ProtCOVER does not produce more informative sets; it picks, per test family, the point on the same coverage-size curve that keeps both classes at the target.

A label-based shift defeats every method, and the weights do not see it. In S5, every family-level method falls to a worse-class coverage of 0.778–0.819, and differences between methods are within noise (Table 2). The label-free density ratio barely departs from one: the effective sample size of the calibration weights is 0.98 of the number of families, as in S1 (0.99), against 0.25 in S2 at $\beta = 2$. When predictor accuracy shifts without a change in the descriptors, the weights neither correct nor flag the shift; a weight diagnostic can show that a shift is visible, never that none exists.

Deployment-realistic features, no alignments, stricter hold-outs. With the raw score only and no variant-count descriptor, Mondrian split still covers only 0.823–0.873 of pathogenic variants per family in S1, and the ranking under shift is unchanged (mean shortfall 0.001 for ProtCOVER, 0.007 for GWCP, 0.019 for HCP), with smaller sets (ProtCOVER 1.29). Dropping alignment depth as well, the descriptor that drives the S2 correction, leaves maximum identity, embedding distance and length, as for a protein without an alignment: every method falls further short, but ProtCOVER's mean shortfall is 0.011 (0.017 in S2, 0.002 in S3 and S4), against 0.019 for GWCP and 0.019 for HCP, smaller than GWCP's by more than the margin in 4 cells and larger in none. On S4s, ProtCOVER's worse-class coverage is at least 0.899 in every cell (mean shortfall 0.000, GWCP 0.006, HCP 0.008). Under the alternative family definitions, ProtCOVER's mean shortfall is 0.002, 0.002 and 0.000, against 0.009, 0.009 and 0.009 for GWCP and 0.023, 0.020 and 0.024 for HCP.

What pooling, the features and the ratio's share contribute. Conditional calibration on one random variant per family, which already attains family-weighted coverage (Section 5), falls short by 0.002 on average over the shifted cells (ProtCOVER 0.001) and varies slightly more across seeds, with the same set size at matched coverage: pooling buys modest stability, not a better trade-off. Richer feature classes are more conservative (with the descriptors added to $(1, \hat{w})$, shortfall 0.000 at set size 1.380, against 1.357), and fitting the ratio on 20% or 50% of the calibration families instead of a third moves ProtCOVER's shortfall by less than the margin in every cell (Appendix Table 5).

External temporal evaluation. All four pre-registered hypotheses were supported in the primary and the secondary analysis (`docs/external_results.md`). On the new variants, Mondrian split covers 0.873 [0.864, 0.883] of pathogenic variants per family in S1 (H1), HCP covers 0.939 and 0.918 (H2), and abstention leaves the error of retained singleton sets unchanged (H4, whose decision rule we fixed after the run, so it is exploratory). H3 holds, but the external data do not discriminate between the shift-aware methods: GWCP, ProtCOVER and both of its variants meet the target in every shifted setting, and only HCP falls short (mean shortfall 0.004, worst 0.880). Coverage is above the target almost everywhere: ESM-1b separates the new variants better than the benchmark variants of the same proteins (per-protein mean AUC 0.930 against 0.899), as expected if recent classifications used computational evidence (PP3/BP4), a caveat we stated in advance. At matched coverage, set sizes are again close (1.150 for ProtCOVER, 1.141 for GWCP, 1.167 for HCP).

The support gate preserves coverage but does not improve selective risk. Abstaining first on the families with the lowest ESS lowered the area under the risk-coverage curve (AURC) relative to random abstention in the majority of seeds in 11 of 30 cells and raised it in 13 (Appendix Fig. 4). Support descriptors locate the families whose coverage needs correcting, which conditional calibration already does; once coverage is controlled they do not predict where the predictor errs, so the gate is not among ProtCOVER's benefits. In ablations, reweighting variants instead of families leaves a mean shortfall of 0.122 (Appendix Table 4); in simulations with a ratio affine in the descriptor, ProtCOVER-D's worse class attains 0.905 where HCP's falls to 0.859 (Appendix C).

7 Limitations

What the guarantees assume. Exact coverage needs exchangeable families (A1), or a family-level shift whose density ratio is a nonnegative function in the span of the features; with an estimated ratio it needs the ratio to be fitted on families independent of those it calibrates (Lemma 1), and the class-conditional versions hold for families that contain the class. Every guarantee is marginal over families: ProtCOVER certifies no particular protein, and the coverage of a small family is dominated by binomial noise. Under any other shift, Corollary 1 bounds the loss by total-variation distances that cannot be estimated from unlabeled data, and when a family's scores and labels given its descriptors differ between calibration and deployment (A2 fails, as in S5), every method under-covers and the ratio gives no warning.

What is only measured. The comparison with the baselines, the robustness to the family definition, the base model and missing alignments, and the external results are empirical. Beyond the 0.01 margin, ProtCOVER's advantage over group-weighted conformal prediction is confined to descriptor shift (S2) on one benchmark whose ten seeds resplit one pool of families; on orphan families, held-out embedding regions and the external evaluations it matches group weighting. The advantage is in coverage, not efficiency: ProtCOVER buys its smaller shortfall with larger sets under shift.

Scope. The ratio-based version needs a batch of unlabeled target families, or a reference panel such as the unlabeled proteome for a single query; ProtCOVER-D does not, at a somewhat larger shortfall. The base model is deliberately simple, and a stronger one may give smaller sets. ProtCOVER outputs prediction sets, not ACMG/AMP evidence strengths; mapping sets to PP3/BP4 strengths is future work.

8 Conclusion

Calibrating a variant effect predictor variant by variant lets well-studied families set its error rate: on ProteinGym, Mondrian split conformal covers only 0.749–0.790 of pathogenic variants per family for four of five predictors, without any shift. Treating families as the exchangeable unit removes this failure, and conditioning a family-weighted quantile regression on a cross-fitted density ratio keeps coverage exact both under family exchangeability and under the estimated shift. The empirical gain is specific to descriptor shift. Under it (S2), ProtCOVER's mean worse-class shortfall is 0.001, against 0.009 for group weighting and 0.032 for HCP; on orphan families and held-out embedding regions it matches group weighting (0.001 against 0.004). The gain is in coverage, not in set size, and a shift in predictor accuracy that the descriptors do not encode defeats every label-free method.

Code Availability

Code and scripts that regenerate every number, table and figure from the public ProteinGym release: `https://github.com/wael-fahmy/protcover` (MIT license).

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A Proofs

Throughout, $F_k = N_k^{-1}\sum_{i=1}^{N_k}\delta_{S_{k,i}}$ is the score distribution of family k (for the test family it involves unobserved labels and is used only in the analysis), and $\gamma_{K+1}(C) = F_{K+1}((-\infty,\hat{q}])$ because a test variant is covered iff its score is at most $\hat{q}$.

Comparison step for (2). Fix weights $p_1,\dots,p_{K+1}$ and define the oracle measure $\mu = \sum_{k=1}^{K+1} p_k F_k$ and $q = \mathrm{Quant}_{1-\alpha}(\mu)$. The computable $\hat{\mu}$ moves the mass p_{K+1} from F_{K+1} to $+\infty$, so $\hat{\mu}((-\infty,u]) \le \mu((-\infty,u])$ for every u, hence $\hat{q} \ge q$ and $\gamma_{K+1}(C) \ge F_{K+1}((-\infty,q])$. Also $\mu((-\infty,q]) \ge 1-\alpha$ by right-continuity.

Proof (Proposition 1). With $\omega \equiv 1$, $p_k = 1/(K+1)$ and q is a symmetric function of $(G_1,\dots,G_{K+1})$. By exchangeability, $\mathbb{E}[F_{K+1}((-\infty,q])] = \mathbb{E}[(K+1)^{-1}\sum_k F_k((-\infty,q])] = \mathbb{E}[\mu((-\infty,q])] \ge 1-\alpha$, and the comparison step gives $\mathrm{Cov}(C) \ge 1-\alpha$.

Proof (Theorem 1). Let $L^o(b) = \sum_{k=1}^{K+1} N_k^{-1}\sum_i \rho_\tau(S_{k,i} - \Phi(D_k)^\top b)$ be the oracle objective, in which the test family enters with its true scores. It is convex, piecewise linear and nonnegative, so its minimizers form a nonempty closed convex polyhedron; let $\tilde{b}$ be the one of minimum Euclidean norm (a measurable function of the data, for example as the limit of the ridge-penalized minimizers as the penalty vanishes; only its symmetry is used), $\tilde{g} = \Phi^\top\tilde{b}$, and $\tilde{\gamma}_k = N_k^{-1}\sum_i \mathbf{1}\{S_{k,i} \le \tilde{g}(D_k)\}$.

(a) First-order condition. Let $f = \Phi^\top b_f$ with $f(D) \ge 0$ P-a.s.; then $f(D_k) \ge 0$ for every k almost surely, which is all the argument uses. The right derivative of $t \mapsto \rho_\tau(S - \tilde{g}(D) - tf(D))$ at $t=0$ is $f(D)(\mathbf{1}\{S \le \tilde{g}(D)\} - \tau)$ (if $S = \tilde{g}(D)$, the residual is $-tf(D) \le 0$ for $t > 0$). The right derivative of $t \mapsto L^o(\tilde{b} + tb_f)$ at 0 is therefore $\sum_{k=1}^{K+1} f(D_k)(\tilde{\gamma}_k - \tau)$, which is nonnegative at a minimizer.

(b) Symmetry. L^o, and hence $\tilde{b}$, is invariant to permutations of the families, so the pairs $(f(D_k),\tilde{\gamma}_k)$, $k \le K+1$, are exchangeable under P^{K+1} and $\mathbb{E}_{P^{K+1}}[f(D_{K+1})(\tilde{\gamma}_{K+1}-\tau)] = (K+1)^{-1}\mathbb{E}\sum_k f(D_k)(\tilde{\gamma}_k-\tau) \ge 0$; the expectation exists because $\tilde{\gamma} \in [0,1]$ and $\mathbb{E}_P f < \infty$.

(c) Change of measure. For any bounded measurable H of the families, $\mathbb{E}_{P^{K+1}}[f(D_{K+1})H] = \mathbb{E}_P f(D) \cdot \mathbb{E}_{P^K\times Q_f}[H]$. With $H = \tilde{\gamma}_{K+1} - \tau$, (b) gives $\mathbb{E}_{P^K\times Q_f}[\tilde{\gamma}_{K+1}] \ge \tau$.

(d) The computable threshold dominates the oracle. We show $\hat{q} \ge \tilde{g}(D_{K+1})$, which gives $\gamma_{K+1}(C) \ge \tilde{\gamma}_{K+1}$ and, with (c), the theorem. The argument follows Proposition 6 of [16] (imputing an upper bound M), with

the supremum over the solution set in place of a min-norm solution, and $M \to \infty$. Write $\phi_t = \Phi(D_{K+1})$. If L in (3) is unbounded below, $\hat{q} = +\infty$. Otherwise $A = \arg\min L$ is nonempty (a polyhedral function bounded below attains its minimum). Pick M larger than every test score and than $\inf_{b\in A} \phi_t^\top b$, and let L^M be L^o with every test score replaced by M. Since $\rho_\tau(u) \geq \tau u$ with equality for $u \geq 0$, $L^M(b) = L(b) + \tau\phi_t^\top b + \rho_\tau(M - \phi_t^\top b) \geq L(b) + \tau M$, with equality when $\phi_t^\top b \leq M$, which holds for some $b \in A$; hence $\min L^M = \min L + \tau M$ and $\arg\min L^M \subseteq A$. Let $b' \in \arg\min L^M$, $a = \tilde{g}(D_{K+1})$ and $a' = \phi_t^\top b'$. Adding $L^o(\tilde{b}) \leq L^o(b')$ and $L^M(b') \leq L^M(\tilde{b})$, the calibration terms cancel and $\sum_i [h_i(a) - h_i(a')] \leq 0$, where $h_i(c) = \rho_\tau(S_{K+1,i} - c) - \rho_\tau(M - c)$ is nondecreasing in c because $S_{K+1,i} \leq M$ and ρ_τ is convex. If $a' < a$, then $h_i(a') \leq h_i(a)$ for every i, so both optimality inequalities hold with equality and $\tilde{b} \in \arg\min L^M \subseteq A$. In either case $\hat{q} = \sup_{b\in A} \phi_t^\top b \geq a$.

The case $f \equiv 1$ gives the statement under A1 (for exchangeable rather than i.i.d. families, (b) applies directly with $f \equiv 1$).

Computing the supremum. Let L' be L with the test term's weight raised from τ to $\tau' > \tau$, so $L'(b) = L(b) - (\tau' - \tau)\phi_t^\top b$. If $b \in \arg\min L$ and $b' \in \arg\min L'$, adding $L(b) \leq L(b')$ and $L'(b') \leq L'(b)$ gives $(\tau' - \tau)\,\phi_t^\top (b' - b) \geq 0$, so every minimizer of L' has $\phi_t^\top b' \geq \hat{q}$. If L is unbounded below along a direction d, its recession slope along d is negative, which forces $\phi_t^\top d > 0$, so L' is unbounded along d as well. If L is bounded but $\hat{q} = +\infty$, then A contains points with $\phi_t^\top b \to \infty$ and L' is unbounded below on A. In every case the threshold computed with $\tau' = \tau + 10^{-4}$ (the perturbed program's optimal fitted value, or $+\infty$ when its dual is infeasible) is at least $\hat{q}$; it equals $\hat{q}$ when $\tau' - \tau$ is small enough for the data (the optimal value is concave and piecewise linear in τ', with finitely many breakpoints, and its right derivative at τ is $-\hat{q}$), and is otherwise conservative. Appendix B checks the computed thresholds against an independent computation under solver tolerances. By Lemma 2 the computed sets keep every guarantee. Standardizing the descriptors (ProtCOVER-D) is an invertible affine reparametrization of b that does not change $\mathcal{F}$, so fitted values, and $\hat{q}$, are unchanged.

Proof (Corollary 1). Let $\phi(g) \in [0, 1]$ be the expected per-family coverage of a test family g over $G_1, \ldots, G_K \sim P$. For $f \in \mathcal{F}$, $f \geq 0$, Theorem 1 gives $\mathbb{E}_{Q_f}\phi \geq 1 - \alpha$, and $|\mathbb{E}_Q\phi - \mathbb{E}_{Q_f}\phi| \leq d_{\mathrm{TV}}(Q, Q_f)$. For $\Phi = (1, \hat{w})$, condition on $\hat{w}$: $f \equiv 1$ and $f = \hat{w}$ are nonnegative members of $\mathcal{F}$ with $Q_1 = P$ and $Q_{\hat{w}} = \hat{Q}$.

Proof (Proposition 2). Under A2 the joint law of $(G_1, \ldots, G_{K+1})$ has density $\prod_{k\leq K} \pi(g_k) \cdot w(d_{K+1})\pi(g_{K+1})$ with respect to a product base measure, where π is a density of P, so the sequence is weighted exchangeable with weight functions $1, \ldots, 1, w$ [38, Def. 1, Lemma 2]. Let E be the event that the unordered collection $\{G_1, \ldots, G_{K+1}\}$ equals $\{g_1, \ldots, g_{K+1}\}$. As in the proof of Lemma 3 of [38], $\mathbb{P}(G_{K+1} = g_k \mid E) = w(d_k)/\sum_j w(d_j) = p_k$ (summing over coincident values). The p_k, μ and q are functions of E, so $\mathbb{E}[F_{K+1}((-\infty, q]) \mid E] = \mu((-\infty, q]) \geq 1 - \alpha$, and the comparison step gives the first claim. For $\hat{w}$ independent of the families, the procedure is invariant to rescaling the weights, so conditionally on $\hat{w}$ it is the procedure of the first claim for the shift $\hat{Q}$; the total-variation argument of Corollary 1 gives the second claim.

Proof (Lemma 1). The ratio for fold j is fitted on the fitting calibration families and the target families outside fold j, rescaled to mean one over the fitting calibration families and clipped by a fixed rule; no other family enters it. Under A2 all families are mutually independent, so $\hat{w}_j$ is independent of the calibration families used in (2) and (3) and of each family of fold j, which, conditionally on $\hat{w}_j$, are still i.i.d. P and Q. Each query family of fold j plays the role of G_{K+1} in Corollary 1 and Proposition 2. If the non-training families are exchangeable (a random split of an exchangeable pool, given the training families), then conditionally on the multiset of families that fit $\hat{w}_j$, the other calibration families together with any one family of fold j are exchangeable, and $\hat{w}_j$, hence Φ, is fixed; step (b) of the proof of Theorem 1 with $f \equiv 1$ applies directly to them, and (d) is unchanged. A unit test in the released code checks that a family's weight does not change when the other calibration families change.

Proof (Proposition 3). Fix y and let $\mathcal{I}_y = \{k \leq K + 1 : N_{k,y} > 0\}$. Conditionally on $\mathcal{I}_y$, with $K + 1 \in \mathcal{I}_y$, the families indexed by $\mathcal{I}_y$ are exchangeable under A1 (the event $\{\mathcal{I}_y = I\}$ is invariant under permutations of I) and independent in the other settings, with laws $P_y = P(\cdot \mid N_y > 0)$ for calibration families and the test law conditioned on $N_y > 0$. Because $\mathbf{1}\{N_y > 0\}$ is a function of G, a test law with density $f(D)/\mathbb{E}_P f$ with respect to P has, after conditioning, density proportional to $f(D)$ with respect to P_y. The label-y procedures

are the procedures above applied to these families with F_k replaced by the label-y score distribution F_k^y (weights $1/N_{k,y}$), so every proof applies verbatim, and the normalizing constants cancel.

Proof (Lemma 2). $\mathbf{1}\{Y \in C'(X)\} \geq \mathbf{1}\{Y \in C(X)\}$ pointwise, so every average and expectation of the indicators is at least as large for C'.

B Protocol Details

Predictors. The five unsupervised predictors are all the non-clinical models in ProteinGym v1.3's clinical score files (columns ESM1b, EVE, GEMME, TranceptEVE_L, PoET; PoET is the 200M model). The other columns are clinical predictors trained or tuned on clinical labels.

Families and homology. MMseqs2 release 18-8cc5c. Clusters: `cluster` with minimum identity 0.3 (0.5 for the sensitivity check), coverage 0.8 and coverage mode 0 (both sequences). Homology search: all against all, sensitivity 7.5, $E \leq 10$, at most 500 hits per query; hits are then filtered at $E \leq 10^{-3}$. Homology components additionally require identity ≥ 0.3 and coverage ≥ 0.5 of both sequences; the strictest definition takes connected components of all hits with $E \leq 10^{-3}$ (1,024 families, 778 single proteins, one component of 867 proteins).

Descriptors. Per protein: maximum identity to any training protein over hits with $E \leq 10^{-3}$ (0 without a hit); $\log(1 + N_{\text{eff}})$ with N_{eff} the sum of ProteinGym's MSA sequence weights; log of N_{eff} divided by alignment length; cosine distance of the mean-pooled last-layer (layer 12) embedding of `esm2_t12_35M_UR50D` to the nearest training protein (sequences over 1022 residues are embedded in windows of 1022 with stride 511 and residue representations averaged over windows); log length. Per family: the maximum of maximum identity and the minimum of embedding distance over member proteins, the mean of the other protein descriptors, and log of the family's number of scored variants. Descriptors are recomputed for every split because they depend on its training families.

Splits. S2 uses z, the mean of the standardized maximum identity and $\log N_{\text{eff}}$ over non-training families, and Efraimidis–Spirakis keys $\log(u)\, e^{\beta z}$ with uniforms u shared across β, so the sweep is coupled and $\beta = 0$ is S1. S3 draws test families uniformly from orphan families and training and calibration families uniformly from the rest. S4 regions are fixed once, before any S4 run: each family's embedding is the mean of its members' unit-normalized ESM2 embeddings, renormalized; after centering, the first 50 principal components are clustered by k-means ($k = 20$, ten restarts, seed 0). Each seed permutes the regions and adds whole regions to the test set until it holds 30% of families (the last region is split at random); training and calibration families are drawn uniformly from the remaining families. S4s permutes the regions in the same way but adds a region only if the test set stays within 30% of families, skipping the others, so that every test region is held out whole (590–624 test families per seed); training and calibration families are drawn uniformly from the remaining families in the proportions 40:30. S5 keeps the S1 training families; z is the standardized mean over the five predictors of each family's within-family AUC (families without both classes get the median), and test families are drawn as in S2 with $\beta = 1$. The ten seeds resplit one fixed pool of families, so they are not independent samples of the population of families.

Base model and conformal methods. The base model standardizes its two features (clipped raw score and the score standardized over the protein's scored variants) and fits L_2-regularized logistic regression ($C = 1$). Dunn et al.'s repeated subsampling uses 50 subsamples. GWCP uses nine bins: maximum identity zero, at most the median of the positive values, or above it, crossed with $\log N_{\text{eff}}$ terciles over calibration and test families; its target proportions are the bin frequencies over all test families (label-free), and it pools at the family level like HCP. For $\hat{w}$, one third of the calibration families fit the classifier (logistic regression with ten candidate penalties, five-fold cross-validated log loss) and the test families are split into five folds. ProtCOVER uses the features $(1, \hat{w}_j)$ on the remaining calibration families for queries of fold j; ProtCOVER-D uses an intercept and maximum identity to training, $\log N_{\text{eff}}$, $\log N_{\text{eff}}$ per position and embedding distance, standardized over all calibration families. Both feature classes were fixed in the configuration before the conditional methods were first run. The alignment-free run, with the deployment-realistic base model, removes $\log N_{\text{eff}}$ and $\log N_{\text{eff}}$

per position from the density ratio, from ProtCOVER-D's features and from the gate, and GWCP crosses maximum identity with embedding-distance terciles instead. The linear programs are solved with HiGHS 1.11 (`highspy`), warm-started across test families, with a cold SciPy solve as fallback.

Threshold audit. The production code solves the perturbed dual with HiGHS's default feasibility tolerances (10^{-7}) and puts a label in the set when its score is at most the threshold plus 10^{-9}, a guard against rounding that can only enlarge sets (Lemma 2). To check that the computed thresholds are conservative under these tolerances, `scripts/audit_thresholds.py` recomputes 1,800 of them independently (ProtCOVER and ProtCOVER-D; seed 0; S1, S2 at $\beta \in \{0.5, 1, 2\}$, S3 and S4; all five predictors; both classes; three random test families per fold): on the primal program with the exact test weight τ, it minimizes L and then maximizes the query's fitted value subject to $L(b) \le L^\star$, both with tolerances of 10^{-10}. Every reference computation succeeded, 16 of them only after rounding forced a relative slack of 10^{-12} on $L^\star$. The two computations agree on whether the threshold is infinite (both are, in 4 cases). The production threshold exceeds the reference by more than 10^{-6} in 7 cases, by at most 2.2×10^{-3}; there $\tau + 10^{-4}$ crosses a breakpoint of the optimal value, because with $\tau + 10^{-5}$ or $\tau + 10^{-6}$ the same thresholds match the reference to 4.4×10^{-9}. It is below the reference in 16 cases, by at most 6.4×10^{-8}; these are exactly the cases that needed slack, which can raise the reference's fitted value, so they are within the precision of the check. Elsewhere the two agree to 2.1×10^{-9}. No audited test variant changes membership between the two thresholds (0 lost, 0 added).

External test set. ClinVar `variant_summary` (October 2026) and `hgvs4variation` map VariationIDs to protein changes in the benchmark's RefSeq proteins; we keep single-nucleotide missense variants on GRCh38 with at least one review star, labeled pathogenic/likely pathogenic or benign/likely benign, whose VariationID is absent from the June 2023 release and whose protein change is not in the benchmark. Excluded as a precaution: VariationIDs present in the July 2023 release and changes that were in the June 2023 release under another VariationID. ESM-1b scores are the precomputed log-likelihood ratios of Brandes et al. [7], oriented as in the benchmark; for proteins of at most 1,022 residues they equal the benchmark's to rounding, so only those are used (longer proteins hold 43% of the new variants). The build script, the pre-registration and its amendment, and the outcome are in the repository.

Gate. The gate acts on the family of a query protein, so all variants of that family share one decision. With standardized support descriptors $\tilde{d}$, the kernel weights $\kappa_{tk} = \exp(-\|\tilde{d}_k - \tilde{d}_t\|^2/2b^2)$ give the effective number of calibration families that resemble query family t, $\mathrm{ESS}_t = (\sum_k \kappa_{tk})^2 / \sum_k \kappa_{tk}^2$, and the test-point mass $\pi_t = p_{K+1}$ of ProtCOVER-W measures how much of the weighted quantile's mass sits on the query itself. The gate abstains when $\mathrm{ESS}_t < m$ or $\pi_t > \pi_{\max}$: the first catches queries with few similar calibration families, the second queries that the ratio marks as far more typical of the targets than of the calibration families. Tuned under the 10% budget (a cap), it abstained on at most 1.2% of test families except at $\beta = 2$ (0.2%–3.6%).

Gate tuning. An inner split of each seed's training families holds out 40% as inner training families (descriptors are recomputed relative to them) and draws 30% as inner test families with the S2 sampler at $\beta = 1$; the rest are inner calibration families. Grids: kernel scale $\{0.1, 0.2, 0.35, 0.5, 1\}$ times the median pairwise distance of standardized support descriptors, $m \in \{0, 2, 5, 10, 20, 40\}$, $\pi_{\max} \in \{1, 0.05, 0.02, 0.01, 0.005\}$, abstention budget 10%, coverage tolerance 0.005.

Metrics. Coverage metrics average per-family coverage over families; set size and singleton rate average over variants. The under-covered fraction uses overall per-family coverage (both classes) for families with at least ten variants, for marginal and class-conditional methods alike; its binomial reference is the expected fraction under Binomial$(N_k, 1-\alpha)$. Bootstrap intervals use 1000 resamples of test families within each seed, holding that seed's calibration draw fixed, so they are conditional on the splits; seed-to-seed variation is reported separately. Random abstention in risk-coverage curves is averaged over 100 random family orders per cell.

C Synthetic Validation

We simulate families with a descriptor $d \sim N(0, 1)$, random offsets, and slopes on the score whose mean depends on d, with a fixed base model, so that the truth is known (300 replicates, Monte Carlo standard error

of every coverage at most 0.002). Under A1 with larger families being easier, variant-level split conformal has variant-weighted coverage 0.900 but family-weighted coverage 0.831, while HCP attains 0.906 and single subsampling 0.902. Under exact family-level covariate shift (A2, test descriptors $N(-\beta, 1)$), HCP drops to 0.826 at $\beta = 1$ and 0.734 at $\beta = 2$; weighting with the true ratio attains 0.911 and 0.950, and with cross-fitted estimated ratios 0.916 and 0.951 (class-conditional pathogenic: 0.916 and 0.947). The excess at $\beta = 2$ is the price of the test-point mass, which grows when few calibration families resemble the test family.

For the class-conditional versions we report the worse of the two class-wise coverages (target 0.90). Under the Gaussian shift at $\beta = 1$ and $\beta = 2$, it is 0.818 and 0.714 for HCP; conditional calibration on $(1, w)$ with the true ratio, the case Theorem 1 covers exactly, attains 0.909 and 0.922; ProtCOVER, with the cross-fitted ratio, 0.913 and 0.939; ProtCOVER-D, whose features $(1, d)$ do not span this ratio, 0.906 and 0.921; and reweighting with the true ratio 0.910 and 0.948. To test the theorem where the descriptor features are exact, a second shift draws calibration descriptors from $U(-2, 2)$ and test descriptors with density $(2 - d)/8$ on $[-2, 2]$, so that $w(d) = (2 - d)/2$ is affine in d: HCP attains 0.859, and ProtCOVER-D 0.905, conditional calibration on the true ratio 0.905, ProtCOVER 0.906 and reweighting with the true ratio 0.904. The central checks are also unit tests in the released code.

D Additional Results

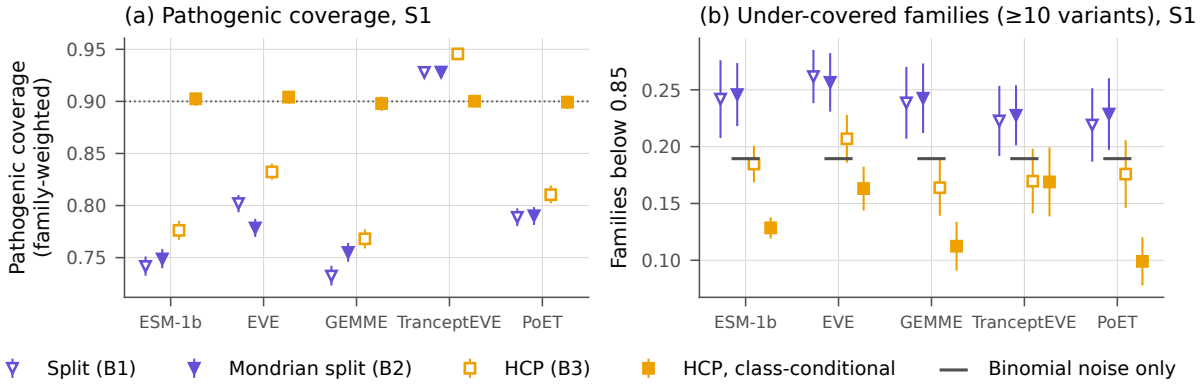


Fig. 1. Exchangeable families (S1, $\alpha = 0.1$). (a) Family-weighted pathogenic coverage Cov_1 with 95% family-bootstrap intervals; the dotted line is the target. (b) Fraction of families with at least ten variants whose coverage is below 0.85 (mean and s.d. over seeds); the gray dash is the expected fraction if every family had coverage exactly 0.90.

Pooling, feature classes and the ratio's share. Over the shifted cells, conditional calibration on one random variant per family and class falls short by 0.002 on average (worst cell 0.016), against 0.001 (0.010) for ProtCOVER, which pools every variant. ProtCOVER's worse-class coverage is higher with a paired interval above zero in 7 cells and lower in 1, its pathogenic coverage varies less across seeds (s.d. 0.018 against 0.020), and both need the same set size at matched coverage (1.332 and 1.329); beyond the margin, pooling helps in 1 cell. Adding the descriptors to $(1, \hat{w})$ removes the shortfall (0.000, worst 0.003) at a larger set size (1.380 against 1.357), and quadratic descriptor features lower ProtCOVER-D's shortfall from 0.003 to 0.001. Fitting the ratio on 20% or 50% of the calibration families instead of a third gives ProtCOVER a shortfall of 0.002 and 0.001, within the margin of the default in every cell (Table 5).

Tables 6–9 give paired differences between ProtCOVER and HCP, GWCP and ProtCOVER-W (class-conditional versions, $\alpha = 0.1$, mean over seeds) with 95% paired family-bootstrap intervals, which are conditional on the splits. Singleton error is the fraction of single-label sets that exclude the true label.

Tables 10–12 report every method, marginal (cc = no) and class-conditional (cc = yes), averaged over the five predictors: family-weighted coverage Cov, variant-weighted coverage, Cov_1, Cov_0, mean set size,

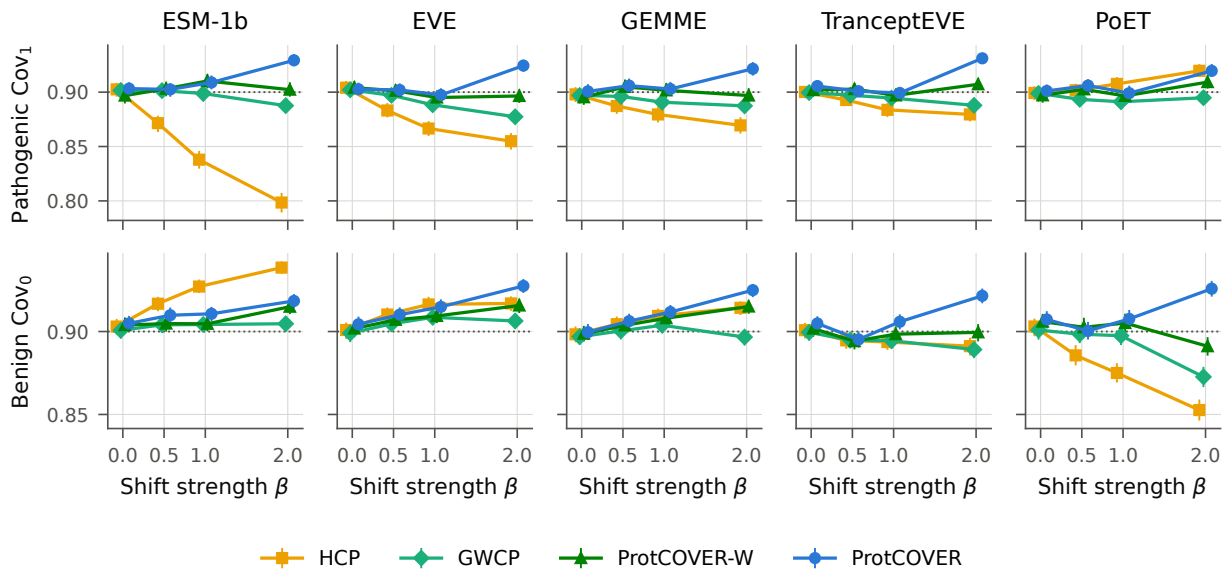


Fig. 2. Class-conditional coverage of both classes under increasing descriptor shift (S2; $\beta = 0$ is S1), $\alpha = 0.1$, with 95% family-bootstrap intervals; methods are offset slightly in β. Mondrian split conformal, far below target, is omitted (Fig. 1).

singleton rate, the fraction of families with at least ten variants whose coverage is below $1 - \alpha - 0.05$, and the fraction of test families on which the method abstained. Per-predictor values with bootstrap intervals, and the summaries of every sensitivity, stress and external run, are in `results/` of the code repository.

Table 10: All methods at $\alpha = 0.1$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.882	0.899	0.798	0.884	1.143	0.857	0.237	0.000
	Split (B1/B2)	yes	0.881	0.898	0.800	0.881	1.142	0.858	0.240	0.000
	HCP (B3)	no	0.901	0.918	0.827	0.903	1.204	0.796	0.180	0.000
	HCP (B3)	yes	0.922	0.929	0.901	0.901	1.304	0.696	0.134	0.000
	Subsample once (B3)	no	0.900	0.918	0.828	0.902	1.204	0.796	0.180	0.000
	Subsample once (B3)	yes	0.922	0.929	0.901	0.900	1.305	0.695	0.136	0.000
	Repeated subsample (B3)	no	0.901	0.918	0.826	0.903	1.205	0.795	0.180	0.000
	Repeated subsample (B3)	yes	0.923	0.929	0.901	0.901	1.305	0.695	0.134	0.000
	GWCP (B4)	no	0.899	0.917	0.824	0.901	1.199	0.801	0.185	0.000
	GWCP (B4)	yes	0.921	0.928	0.900	0.899	1.300	0.700	0.136	0.000
	ProtCOVER	no	0.903	0.914	0.828	0.906	1.195	0.804	0.185	0.000
	ProtCOVER	yes	0.923	0.928	0.903	0.904	1.306	0.694	0.140	0.000
	ProtCOVER-D	no	0.906	0.907	0.827	0.912	1.180	0.820	0.186	0.000
	ProtCOVER-D	yes	0.920	0.923	0.904	0.906	1.297	0.703	0.165	0.000
	ProtCOVER-W	no	0.900	0.918	0.826	0.903	1.204	0.796	0.181	0.000
	ProtCOVER-W	yes	0.923	0.930	0.899	0.902	1.303	0.697	0.133	0.000
	(i)+(iii) gate	no	0.901	0.918	0.827	0.903	1.204	0.796	0.180	0.000
	(i)+(iii) gate	yes	0.923	0.929	0.901	0.901	1.304	0.696	0.134	0.000
	ProtCOVER-W + gate	no	0.901	0.918	0.826	0.903	1.204	0.796	0.180	0.000
	ProtCOVER-W + gate	yes	0.923	0.930	0.899	0.903	1.304	0.696	0.133	0.000
	Support-only abst. (B5)	no	0.901	0.918	0.826	0.903	1.204	0.796	0.181	0.000
	Support-only abst. (B5)	yes	0.923	0.930	0.899	0.902	1.304	0.696	0.133	0.000
	Random abst. (B5)	no	0.901	0.918	0.826	0.903	1.204	0.796	0.180	0.000
	Random abst. (B5)	yes	0.923	0.930	0.899	0.902	1.304	0.696	0.133	0.000

All methods at $\alpha = 0.1$, continued

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S2, $\beta = 1$	Split (B1/B2)	no	0.875	0.893	0.772	0.883	1.137	0.863	0.252	0.000
	Split (B1/B2)	yes	0.873	0.891	0.758	0.889	1.137	0.863	0.255	0.000
	HCP (B3)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	HCP (B3)	yes	0.917	0.925	0.875	0.904	1.291	0.709	0.147	0.000
	Subsample once (B3)	no	0.893	0.912	0.800	0.902	1.195	0.805	0.200	0.000
	Subsample once (B3)	yes	0.918	0.926	0.877	0.904	1.294	0.706	0.146	0.000
	Repeated subsample (B3)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	Repeated subsample (B3)	yes	0.917	0.925	0.875	0.904	1.290	0.710	0.147	0.000
	GWCP (B4)	no	0.896	0.914	0.805	0.903	1.202	0.798	0.195	0.000
	GWCP (B4)	yes	0.921	0.928	0.893	0.902	1.318	0.682	0.140	0.000
	ProtCOVER	no	0.907	0.923	0.831	0.912	1.245	0.755	0.170	0.000
	ProtCOVER	yes	0.927	0.931	0.901	0.910	1.338	0.662	0.132	0.000
	ProtCOVER-D	no	0.903	0.908	0.807	0.914	1.188	0.812	0.191	0.000
	ProtCOVER-D	yes	0.921	0.924	0.901	0.909	1.327	0.673	0.164	0.000
	ProtCOVER-W	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	ProtCOVER-W	yes	0.925	0.931	0.900	0.905	1.340	0.660	0.130	0.000
	(i)+(iii) gate	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	(i)+(iii) gate	yes	0.917	0.925	0.875	0.904	1.291	0.709	0.147	0.000
	ProtCOVER-W + gate	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	ProtCOVER-W + gate	yes	0.925	0.931	0.900	0.905	1.340	0.660	0.129	0.000
	Support-only abst. (B5)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	Support-only abst. (B5)	yes	0.925	0.931	0.900	0.905	1.341	0.659	0.130	0.000
	Random abst. (B5)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	Random abst. (B5)	yes	0.925	0.931	0.900	0.905	1.341	0.659	0.130	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.874	0.892	0.764	0.883	1.136	0.864	0.255	0.000
	Split (B1/B2)	yes	0.871	0.890	0.746	0.889	1.136	0.864	0.259	0.000
	HCP (B3)	no	0.892	0.911	0.793	0.900	1.192	0.808	0.204	0.000
	HCP (B3)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	Subsample once (B3)	no	0.889	0.909	0.792	0.896	1.187	0.813	0.211	0.000
	Subsample once (B3)	yes	0.912	0.922	0.863	0.901	1.279	0.721	0.159	0.000
	Repeated subsample (B3)	no	0.891	0.911	0.792	0.900	1.192	0.808	0.204	0.000
	Repeated subsample (B3)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	GWCP (B4)	no	0.896	0.915	0.801	0.904	1.212	0.788	0.196	0.000
	GWCP (B4)	yes	0.915	0.923	0.887	0.894	1.315	0.685	0.159	0.000
	ProtCOVER	no	0.927	0.941	0.870	0.928	1.400	0.600	0.131	0.000
	ProtCOVER	yes	0.939	0.944	0.925	0.924	1.458	0.542	0.110	0.000
	ProtCOVER-D	no	0.901	0.910	0.801	0.913	1.197	0.803	0.190	0.000
	ProtCOVER-D	yes	0.921	0.924	0.902	0.909	1.354	0.646	0.166	0.000
	ProtCOVER-W	no	0.909	0.927	0.826	0.914	1.260	0.740	0.161	0.000
	ProtCOVER-W	yes	0.927	0.935	0.902	0.907	1.363	0.637	0.127	0.000
	(i)+(iii) gate	no	0.892	0.911	0.793	0.900	1.192	0.808	0.204	0.000
	(i)+(iii) gate	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	ProtCOVER-W + gate	no	0.909	0.927	0.827	0.914	1.264	0.736	0.161	0.004
	ProtCOVER-W + gate	yes	0.928	0.936	0.904	0.908	1.372	0.628	0.125	0.022
	Support-only abst. (B5)	no	0.909	0.927	0.827	0.914	1.263	0.737	0.160	0.004
	Support-only abst. (B5)	yes	0.928	0.936	0.904	0.909	1.375	0.625	0.125	0.022
	Random abst. (B5)	no	0.909	0.928	0.827	0.915	1.264	0.736	0.160	0.004
	Random abst. (B5)	yes	0.928	0.936	0.904	0.909	1.375	0.625	0.125	0.022
S3 (orphan families)	Split (B1/B2)	no	0.875	0.891	0.789	0.878	1.137	0.863	0.254	0.000
	Split (B1/B2)	yes	0.874	0.890	0.787	0.878	1.135	0.865	0.251	0.000
	HCP (B3)	no	0.897	0.913	0.821	0.899	1.204	0.796	0.193	0.000
	HCP (B3)	yes	0.922	0.931	0.904	0.896	1.308	0.692	0.129	0.000
	Subsample once (B3)	no	0.896	0.912	0.820	0.898	1.204	0.796	0.195	0.000
	Subsample once (B3)	yes	0.922	0.931	0.904	0.896	1.309	0.691	0.128	0.000
	Repeated subsample (B3)	no	0.896	0.912	0.821	0.899	1.203	0.797	0.194	0.000

All methods at $\alpha = 0.1$, continued

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
	Repeated subsample (B3)	yes	0.922	0.931	0.904	0.896	1.308	0.692	0.129	0.000
	GWCP (B4)	no	0.901	0.917	0.827	0.903	1.220	0.780	0.182	0.000
	GWCP (B4)	yes	0.924	0.931	0.898	0.903	1.312	0.688	0.128	0.000
	ProtCOVER	no	0.905	0.921	0.834	0.908	1.236	0.764	0.170	0.000
	ProtCOVER	yes	0.927	0.934	0.902	0.909	1.327	0.673	0.121	0.000
	ProtCOVER-D	no	0.908	0.913	0.829	0.916	1.208	0.792	0.180	0.000
	ProtCOVER-D	yes	0.921	0.922	0.905	0.907	1.314	0.686	0.165	0.000
	ProtCOVER-W	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	ProtCOVER-W	yes	0.928	0.934	0.900	0.910	1.329	0.671	0.120	0.000
	(i)+(iii) gate	no	0.897	0.913	0.821	0.899	1.204	0.796	0.193	0.000
	(i)+(iii) gate	yes	0.922	0.931	0.904	0.897	1.309	0.691	0.129	0.003
	ProtCOVER-W + gate	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	ProtCOVER-W + gate	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004
	Support-only abst. (B5)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	Support-only abst. (B5)	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004
	Random abst. (B5)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	Random abst. (B5)	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004
S4 (held-out regions)	Split (B1/B2)	no	0.878	0.895	0.801	0.882	1.145	0.855	0.245	0.000
	Split (B1/B2)	yes	0.876	0.895	0.802	0.877	1.141	0.859	0.248	0.000
	HCP (B3)	no	0.897	0.915	0.830	0.900	1.205	0.795	0.188	0.000
	HCP (B3)	yes	0.920	0.928	0.904	0.896	1.300	0.700	0.137	0.000
	Subsample once (B3)	no	0.899	0.916	0.835	0.900	1.212	0.788	0.186	0.000
	Subsample once (B3)	yes	0.920	0.928	0.904	0.896	1.301	0.699	0.138	0.000
	Repeated subsample (B3)	no	0.897	0.914	0.829	0.900	1.205	0.795	0.189	0.000
	Repeated subsample (B3)	yes	0.920	0.928	0.904	0.896	1.300	0.700	0.137	0.000
	GWCP (B4)	no	0.897	0.915	0.830	0.901	1.207	0.793	0.188	0.000
	GWCP (B4)	yes	0.920	0.926	0.899	0.899	1.294	0.706	0.141	0.000
	ProtCOVER	no	0.907	0.921	0.842	0.910	1.251	0.749	0.173	0.003
	ProtCOVER	yes	0.927	0.934	0.914	0.906	1.342	0.658	0.129	0.003
	ProtCOVER-D	no	0.908	0.910	0.836	0.916	1.204	0.796	0.182	0.000
	ProtCOVER-D	yes	0.920	0.922	0.902	0.911	1.311	0.689	0.167	0.000
	ProtCOVER-W	no	0.903	0.919	0.836	0.907	1.232	0.768	0.177	0.000
	ProtCOVER-W	yes	0.923	0.926	0.895	0.908	1.312	0.688	0.138	0.000
	(i)+(iii) gate	no	0.897	0.915	0.830	0.901	1.207	0.793	0.188	0.001
	(i)+(iii) gate	yes	0.920	0.928	0.904	0.896	1.304	0.696	0.137	0.003
	ProtCOVER-W + gate	no	0.903	0.919	0.837	0.907	1.233	0.767	0.177	0.001
	ProtCOVER-W + gate	yes	0.923	0.926	0.895	0.909	1.316	0.684	0.137	0.005
	Support-only abst. (B5)	no	0.903	0.919	0.837	0.907	1.232	0.768	0.177	0.001
	Support-only abst. (B5)	yes	0.923	0.926	0.896	0.908	1.315	0.685	0.137	0.005
	Random abst. (B5)	no	0.903	0.919	0.837	0.907	1.232	0.768	0.177	0.001
	Random abst. (B5)	yes	0.923	0.926	0.896	0.908	1.315	0.685	0.137	0.005

Table 11: All methods at $\alpha = 0.05$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.933	0.949	0.876	0.934	1.336	0.664	0.152	0.000
	Split (B1/B2)	yes	0.933	0.949	0.876	0.936	1.335	0.665	0.151	0.000
	HCP (B3)	no	0.950	0.966	0.904	0.953	1.443	0.557	0.095	0.000
	HCP (B3)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.075	0.000
	Subsample once (B3)	no	0.950	0.965	0.902	0.952	1.441	0.559	0.098	0.000
	Subsample once (B3)	yes	0.963	0.967	0.951	0.950	1.515	0.485	0.076	0.000
	Repeated subsample (B3)	no	0.951	0.966	0.904	0.953	1.444	0.556	0.094	0.000
	Repeated subsample (B3)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.076	0.000
	GWCP (B4)	no	0.949	0.964	0.901	0.951	1.432	0.568	0.101	0.000

All methods at $\alpha = 0.05$, continued

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
	GWCP (B4)	yes	0.961	0.966	0.950	0.948	1.507	0.493	0.080	0.000
	ProtCOVER	no	0.953	0.964	0.906	0.956	1.437	0.563	0.097	0.000
	ProtCOVER	yes	0.965	0.969	0.954	0.954	1.532	0.468	0.076	0.000
	ProtCOVER-D	no	0.956	0.960	0.907	0.961	1.418	0.582	0.094	0.000
	ProtCOVER-D	yes	0.963	0.966	0.956	0.956	1.535	0.465	0.090	0.000
	ProtCOVER-W	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	ProtCOVER-W	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	(i)+(iii) gate	no	0.950	0.966	0.904	0.953	1.443	0.557	0.095	0.000
	(i)+(iii) gate	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.075	0.000
	ProtCOVER-W + gate	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	ProtCOVER-W + gate	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	Support-only abst. (B5)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	Support-only abst. (B5)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	Random abst. (B5)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	Random abst. (B5)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.928	0.946	0.859	0.934	1.330	0.670	0.167	0.000
	Split (B1/B2)	yes	0.927	0.944	0.843	0.940	1.325	0.675	0.167	0.000
	HCP (B3)	no	0.946	0.962	0.889	0.950	1.431	0.569	0.108	0.000
	HCP (B3)	yes	0.960	0.967	0.936	0.952	1.516	0.484	0.080	0.000
	Subsample once (B3)	no	0.947	0.963	0.890	0.951	1.441	0.559	0.105	0.000
	Subsample once (B3)	yes	0.958	0.966	0.935	0.949	1.510	0.490	0.084	0.000
	Repeated subsample (B3)	no	0.946	0.962	0.889	0.950	1.432	0.568	0.109	0.000
	Repeated subsample (B3)	yes	0.960	0.967	0.935	0.952	1.515	0.485	0.080	0.000
	GWCP (B4)	no	0.947	0.964	0.892	0.951	1.447	0.553	0.104	0.000
	GWCP (B4)	yes	0.961	0.966	0.944	0.950	1.527	0.473	0.081	0.000
	ProtCOVER	no	0.957	0.970	0.916	0.959	1.505	0.495	0.083	0.000
	ProtCOVER	yes	0.966	0.970	0.952	0.958	1.559	0.441	0.068	0.000
	ProtCOVER-D	no	0.955	0.961	0.897	0.963	1.427	0.573	0.097	0.000
	ProtCOVER-D	yes	0.964	0.967	0.954	0.958	1.556	0.444	0.089	0.000
	ProtCOVER-W	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	ProtCOVER-W	yes	0.965	0.969	0.951	0.955	1.553	0.447	0.069	0.000
	(i)+(iii) gate	no	0.946	0.962	0.889	0.950	1.431	0.569	0.108	0.000
	(i)+(iii) gate	yes	0.960	0.967	0.936	0.952	1.516	0.484	0.080	0.000
	ProtCOVER-W + gate	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	ProtCOVER-W + gate	yes	0.965	0.969	0.951	0.955	1.553	0.447	0.069	0.000
	Support-only abst. (B5)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	Support-only abst. (B5)	yes	0.965	0.969	0.951	0.955	1.554	0.446	0.069	0.000
	Random abst. (B5)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	Random abst. (B5)	yes	0.965	0.969	0.951	0.955	1.554	0.446	0.069	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.927	0.945	0.852	0.933	1.329	0.671	0.170	0.000
	Split (B1/B2)	yes	0.925	0.943	0.833	0.941	1.324	0.676	0.173	0.000
	HCP (B3)	no	0.944	0.961	0.884	0.949	1.427	0.573	0.114	0.000
	HCP (B3)	yes	0.957	0.966	0.929	0.950	1.512	0.488	0.087	0.000
	Subsample once (B3)	no	0.943	0.959	0.878	0.949	1.416	0.584	0.119	0.000
	Subsample once (B3)	yes	0.956	0.964	0.926	0.949	1.505	0.495	0.090	0.000
	Repeated subsample (B3)	no	0.945	0.961	0.883	0.949	1.427	0.573	0.115	0.000
	Repeated subsample (B3)	yes	0.957	0.966	0.929	0.949	1.511	0.489	0.087	0.000
	GWCP (B4)	no	0.946	0.963	0.888	0.950	1.456	0.544	0.109	0.000
	GWCP (B4)	yes	0.956	0.963	0.943	0.942	1.526	0.474	0.096	0.000
	ProtCOVER	no	0.969	0.978	0.940	0.969	1.632	0.368	0.060	0.016
	ProtCOVER	yes	0.975	0.978	0.968	0.967	1.670	0.330	0.052	0.016
	ProtCOVER-D	no	0.955	0.963	0.896	0.963	1.443	0.557	0.096	0.000
	ProtCOVER-D	yes	0.965	0.967	0.956	0.959	1.580	0.420	0.090	0.000
	ProtCOVER-W	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	ProtCOVER-W	yes	0.969	0.974	0.962	0.958	1.612	0.388	0.064	0.000

All methods at $\alpha = 0.05$, continued

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
	(i)+(iii) gate	no	0.944	0.961	0.884	0.949	1.427	0.573	0.114	0.000
	(i)+(iii) gate	yes	0.957	0.966	0.929	0.950	1.512	0.488	0.087	0.000
	ProtCOVER-W + gate	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	ProtCOVER-W + gate	yes	0.969	0.974	0.962	0.958	1.613	0.387	0.064	0.014
	Support-only abst. (B5)	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	Support-only abst. (B5)	yes	0.969	0.974	0.962	0.958	1.617	0.383	0.063	0.014
	Random abst. (B5)	no	0.962	0.975	0.924	0.963	1.572	0.428	0.068	0.000
	Random abst. (B5)	yes	0.969	0.974	0.962	0.958	1.616	0.384	0.063	0.014
S3 (orphan families)	Split (B1/B2)	no	0.929	0.944	0.870	0.931	1.330	0.670	0.159	0.000
	Split (B1/B2)	yes	0.930	0.944	0.865	0.936	1.330	0.670	0.156	0.000
	HCP (B3)	no	0.949	0.963	0.903	0.951	1.448	0.552	0.096	0.000
	HCP (B3)	yes	0.963	0.970	0.952	0.950	1.527	0.473	0.069	0.000
	Subsample once (B3)	no	0.946	0.961	0.899	0.948	1.440	0.560	0.105	0.000
	Subsample once (B3)	yes	0.962	0.970	0.952	0.948	1.525	0.475	0.070	0.000
	Repeated subsample (B3)	no	0.949	0.963	0.903	0.951	1.447	0.553	0.097	0.000
	Repeated subsample (B3)	yes	0.963	0.970	0.952	0.950	1.528	0.472	0.068	0.000
	GWCP (B4)	no	0.951	0.965	0.905	0.954	1.467	0.533	0.091	0.000
	GWCP (B4)	yes	0.964	0.969	0.951	0.953	1.527	0.473	0.069	0.000
	ProtCOVER	no	0.956	0.968	0.914	0.959	1.494	0.506	0.078	0.000
	ProtCOVER	yes	0.967	0.972	0.954	0.958	1.550	0.450	0.063	0.000
	ProtCOVER-D	no	0.958	0.964	0.910	0.964	1.443	0.557	0.090	0.000
	ProtCOVER-D	yes	0.965	0.966	0.956	0.958	1.530	0.470	0.087	0.000
	ProtCOVER-W	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	ProtCOVER-W	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
	(i)+(iii) gate	no	0.949	0.963	0.903	0.951	1.448	0.552	0.096	0.000
	(i)+(iii) gate	yes	0.963	0.970	0.952	0.950	1.527	0.473	0.069	0.000
	ProtCOVER-W + gate	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	ProtCOVER-W + gate	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
	Support-only abst. (B5)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	Support-only abst. (B5)	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
	Random abst. (B5)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	Random abst. (B5)	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
S4 (held-out regions)	Split (B1/B2)	no	0.930	0.947	0.879	0.933	1.340	0.660	0.154	0.000
	Split (B1/B2)	yes	0.931	0.947	0.878	0.932	1.337	0.663	0.155	0.000
	HCP (B3)	no	0.949	0.964	0.907	0.951	1.447	0.553	0.097	0.000
	HCP (B3)	yes	0.961	0.967	0.953	0.946	1.510	0.490	0.077	0.000
	Subsample once (B3)	no	0.948	0.964	0.907	0.950	1.445	0.555	0.099	0.000
	Subsample once (B3)	yes	0.960	0.966	0.952	0.946	1.506	0.494	0.077	0.000
	Repeated subsample (B3)	no	0.949	0.964	0.908	0.951	1.448	0.552	0.096	0.000
	Repeated subsample (B3)	yes	0.961	0.967	0.953	0.946	1.509	0.491	0.077	0.000
	GWCP (B4)	no	0.948	0.963	0.906	0.950	1.442	0.558	0.101	0.000
	GWCP (B4)	yes	0.960	0.965	0.950	0.947	1.503	0.497	0.080	0.000
	ProtCOVER	no	0.956	0.969	0.919	0.959	1.501	0.499	0.084	0.017
	ProtCOVER	yes	0.966	0.970	0.959	0.954	1.559	0.441	0.070	0.017
	ProtCOVER-D	no	0.958	0.963	0.916	0.964	1.458	0.542	0.090	0.000
	ProtCOVER-D	yes	0.964	0.966	0.957	0.957	1.547	0.453	0.091	0.000
	ProtCOVER-W	no	0.953	0.967	0.912	0.956	1.488	0.512	0.089	0.000
	ProtCOVER-W	yes	0.962	0.965	0.948	0.953	1.528	0.472	0.080	0.000
	(i)+(iii) gate	no	0.949	0.964	0.907	0.951	1.447	0.553	0.097	0.000
	(i)+(iii) gate	yes	0.961	0.967	0.954	0.947	1.515	0.485	0.076	0.007
	ProtCOVER-W + gate	no	0.953	0.967	0.912	0.956	1.488	0.512	0.089	0.000
	ProtCOVER-W + gate	yes	0.962	0.965	0.948	0.953	1.528	0.472	0.080	0.002
	Support-only abst. (B5)	no	0.953	0.967	0.912	0.956	1.488	0.512	0.089	0.000
	Support-only abst. (B5)	yes	0.962	0.965	0.948	0.953	1.528	0.472	0.080	0.002
	Random abst. (B5)	no	0.953	0.967	0.912	0.956	1.488	0.512	0.089	0.000

All methods at $\alpha = 0.05$, continued

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
	Random abst. (B5)	yes	0.962	0.965	0.948	0.953	1.529	0.471	0.080	0.002

Table 12: All methods at $\alpha = 0.2$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.786	0.797	0.675	0.787	0.927	0.919	0.282	0.000
	Split (B1/B2)	yes	0.782	0.796	0.680	0.779	0.923	0.921	0.287	0.000
	HCP (B3)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	HCP (B3)	yes	0.840	0.846	0.804	0.803	1.045	0.952	0.164	0.000
	Subsample once (B3)	no	0.800	0.812	0.692	0.803	0.954	0.947	0.243	0.000
	Subsample once (B3)	yes	0.840	0.846	0.802	0.804	1.046	0.951	0.165	0.000
	Repeated subsample (B3)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	Repeated subsample (B3)	yes	0.840	0.847	0.804	0.803	1.045	0.952	0.164	0.000
	GWCP (B4)	no	0.799	0.810	0.692	0.800	0.949	0.947	0.246	0.000
	GWCP (B4)	yes	0.839	0.845	0.803	0.801	1.043	0.953	0.166	0.000
	ProtCOVER	no	0.802	0.808	0.695	0.805	0.947	0.937	0.247	0.000
	ProtCOVER	yes	0.837	0.841	0.801	0.806	1.038	0.950	0.176	0.000
	ProtCOVER-D	no	0.805	0.801	0.694	0.809	0.935	0.920	0.255	0.000
	ProtCOVER-D	yes	0.829	0.830	0.805	0.804	1.029	0.933	0.212	0.000
	ProtCOVER-W	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.000
	ProtCOVER-W	yes	0.840	0.847	0.799	0.805	1.044	0.953	0.163	0.000
	(i)+(iii) gate	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	(i)+(iii) gate	yes	0.840	0.847	0.804	0.803	1.046	0.951	0.164	0.001
	ProtCOVER-W + gate	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	ProtCOVER-W + gate	yes	0.840	0.847	0.800	0.806	1.044	0.953	0.163	0.000
	Support-only abst. (B5)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	Support-only abst. (B5)	yes	0.840	0.847	0.799	0.806	1.044	0.953	0.163	0.000
	Random abst. (B5)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	Random abst. (B5)	yes	0.840	0.847	0.799	0.806	1.045	0.953	0.163	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.777	0.789	0.641	0.788	0.920	0.909	0.305	0.000
	Split (B1/B2)	yes	0.774	0.787	0.630	0.792	0.917	0.914	0.308	0.000
	HCP (B3)	no	0.790	0.803	0.658	0.801	0.942	0.941	0.266	0.000
	HCP (B3)	yes	0.829	0.836	0.754	0.811	1.024	0.967	0.187	0.000
	Subsample once (B3)	no	0.786	0.798	0.653	0.798	0.935	0.933	0.280	0.000
	Subsample once (B3)	yes	0.828	0.836	0.752	0.810	1.023	0.965	0.189	0.000
	Repeated subsample (B3)	no	0.790	0.803	0.659	0.801	0.943	0.941	0.266	0.000
	Repeated subsample (B3)	yes	0.828	0.836	0.754	0.810	1.023	0.967	0.189	0.000
	GWCP (B4)	no	0.794	0.806	0.663	0.804	0.948	0.943	0.259	0.000
	GWCP (B4)	yes	0.835	0.842	0.785	0.804	1.050	0.944	0.179	0.000
	ProtCOVER	no	0.804	0.816	0.680	0.813	0.971	0.933	0.239	0.000
	ProtCOVER	yes	0.843	0.846	0.800	0.814	1.075	0.917	0.163	0.000
	ProtCOVER-D	no	0.800	0.800	0.666	0.812	0.939	0.923	0.264	0.000
	ProtCOVER-D	yes	0.831	0.832	0.797	0.809	1.050	0.922	0.211	0.000
	ProtCOVER-W	no	0.800	0.813	0.672	0.810	0.960	0.950	0.246	0.000
	ProtCOVER-W	yes	0.844	0.850	0.800	0.810	1.075	0.921	0.162	0.000
	(i)+(iii) gate	no	0.790	0.803	0.658	0.801	0.942	0.941	0.266	0.000
	(i)+(iii) gate	yes	0.829	0.836	0.754	0.811	1.025	0.967	0.187	0.001
	ProtCOVER-W + gate	no	0.800	0.813	0.673	0.811	0.962	0.949	0.246	0.001
	ProtCOVER-W + gate	yes	0.844	0.850	0.801	0.811	1.075	0.921	0.162	0.000
	Support-only abst. (B5)	no	0.800	0.813	0.672	0.811	0.962	0.949	0.246	0.001
	Support-only abst. (B5)	yes	0.844	0.850	0.801	0.810	1.075	0.921	0.162	0.000
	Random abst. (B5)	no	0.800	0.813	0.672	0.811	0.962	0.949	0.246	0.001
	Random abst. (B5)	yes	0.844	0.850	0.801	0.810	1.075	0.921	0.162	0.000

All methods at $\alpha = 0.2$, continued

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S2, $\beta = 2$	Split (B1/B2)	no	0.776	0.788	0.633	0.789	0.920	0.907	0.306	0.000
	Split (B1/B2)	yes	0.773	0.786	0.618	0.794	0.917	0.913	0.312	0.000
	HCP (B3)	no	0.790	0.803	0.650	0.803	0.942	0.941	0.268	0.000
	HCP (B3)	yes	0.824	0.833	0.736	0.812	1.018	0.971	0.197	0.000
	Subsample once (B3)	no	0.788	0.801	0.648	0.801	0.939	0.937	0.274	0.000
	Subsample once (B3)	yes	0.825	0.834	0.736	0.813	1.020	0.967	0.196	0.000
	Repeated subsample (B3)	no	0.791	0.803	0.650	0.803	0.942	0.941	0.266	0.000
	Repeated subsample (B3)	yes	0.824	0.833	0.736	0.811	1.018	0.970	0.197	0.000
	GWCP (B4)	no	0.798	0.810	0.661	0.810	0.957	0.941	0.254	0.000
	GWCP (B4)	yes	0.830	0.837	0.778	0.798	1.054	0.931	0.192	0.000
	ProtCOVER	no	0.829	0.842	0.718	0.836	1.064	0.865	0.193	0.000
	ProtCOVER	yes	0.856	0.859	0.833	0.820	1.173	0.820	0.157	0.000
	ProtCOVER-D	no	0.801	0.804	0.659	0.814	0.946	0.929	0.260	0.000
	ProtCOVER-D	yes	0.828	0.830	0.789	0.808	1.068	0.908	0.214	0.000
	ProtCOVER-W	no	0.808	0.821	0.675	0.819	0.979	0.945	0.229	0.000
	ProtCOVER-W	yes	0.842	0.850	0.796	0.810	1.079	0.913	0.165	0.000
	(i)+(iii) gate	no	0.790	0.803	0.650	0.803	0.942	0.941	0.268	0.000
	(i)+(iii) gate	yes	0.825	0.833	0.736	0.812	1.019	0.970	0.196	0.001
	ProtCOVER-W + gate	no	0.811	0.824	0.684	0.821	0.998	0.928	0.225	0.018
	ProtCOVER-W + gate	yes	0.845	0.853	0.800	0.813	1.096	0.897	0.162	0.019
	Support-only abst. (B5)	no	0.811	0.824	0.682	0.822	0.998	0.928	0.225	0.018
	Support-only abst. (B5)	yes	0.845	0.852	0.799	0.813	1.094	0.899	0.162	0.019
	Random abst. (B5)	no	0.811	0.824	0.683	0.822	0.999	0.927	0.225	0.018
	Random abst. (B5)	yes	0.845	0.852	0.799	0.813	1.095	0.898	0.162	0.019
S3 (orphan families)	Split (B1/B2)	no	0.778	0.789	0.664	0.781	0.924	0.912	0.300	0.000
	Split (B1/B2)	yes	0.774	0.788	0.664	0.775	0.919	0.916	0.307	0.000
	HCP (B3)	no	0.794	0.805	0.685	0.797	0.951	0.946	0.257	0.000
	HCP (B3)	yes	0.836	0.846	0.799	0.796	1.040	0.955	0.174	0.000
	Subsample once (B3)	no	0.795	0.806	0.682	0.801	0.954	0.943	0.255	0.000
	Subsample once (B3)	yes	0.835	0.846	0.799	0.794	1.039	0.958	0.176	0.000
	Repeated subsample (B3)	no	0.794	0.806	0.686	0.797	0.951	0.946	0.257	0.000
	Repeated subsample (B3)	yes	0.836	0.846	0.798	0.796	1.040	0.956	0.175	0.000
	GWCP (B4)	no	0.802	0.814	0.695	0.806	0.967	0.957	0.236	0.000
	GWCP (B4)	yes	0.839	0.847	0.792	0.805	1.046	0.948	0.170	0.000
	ProtCOVER	no	0.805	0.816	0.699	0.809	0.972	0.952	0.233	0.000
	ProtCOVER	yes	0.842	0.850	0.796	0.810	1.056	0.939	0.165	0.000
	ProtCOVER-D	no	0.809	0.810	0.699	0.815	0.961	0.934	0.245	0.000
	ProtCOVER-D	yes	0.833	0.833	0.805	0.810	1.050	0.924	0.206	0.000
	ProtCOVER-W	no	0.807	0.819	0.700	0.810	0.976	0.956	0.227	0.000
	ProtCOVER-W	yes	0.843	0.850	0.793	0.812	1.056	0.939	0.163	0.000
	(i)+(iii) gate	no	0.794	0.805	0.685	0.797	0.951	0.946	0.257	0.000
	(i)+(iii) gate	yes	0.837	0.847	0.800	0.797	1.044	0.951	0.173	0.005
	ProtCOVER-W + gate	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	ProtCOVER-W + gate	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003
	Support-only abst. (B5)	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	Support-only abst. (B5)	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003
	Random abst. (B5)	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	Random abst. (B5)	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003
S4 (held-out regions)	Split (B1/B2)	no	0.782	0.790	0.678	0.786	0.926	0.912	0.290	0.000
	Split (B1/B2)	yes	0.778	0.791	0.683	0.773	0.921	0.916	0.296	0.000
	HCP (B3)	no	0.796	0.804	0.696	0.800	0.949	0.946	0.250	0.000
	HCP (B3)	yes	0.837	0.844	0.808	0.795	1.044	0.955	0.168	0.000
	Subsample once (B3)	no	0.796	0.804	0.696	0.800	0.950	0.944	0.250	0.000
	Subsample once (B3)	yes	0.835	0.843	0.809	0.793	1.043	0.954	0.170	0.000
	Repeated subsample (B3)	no	0.796	0.804	0.696	0.800	0.949	0.946	0.250	0.000

All methods at $\alpha = 0.2$, continued

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
	Repeated subsample (B3)	yes	0.837	0.844	0.808	0.796	1.043	0.955	0.168	0.000
	GWCP (B4)	no	0.798	0.806	0.698	0.802	0.953	0.949	0.245	0.000
	GWCP (B4)	yes	0.836	0.842	0.800	0.801	1.040	0.958	0.171	0.000
	ProtCOVER	no	0.806	0.813	0.707	0.810	0.971	0.937	0.235	0.001
	ProtCOVER	yes	0.843	0.851	0.814	0.808	1.068	0.929	0.162	0.001
	ProtCOVER-D	no	0.807	0.801	0.704	0.814	0.946	0.926	0.246	0.000
	ProtCOVER-D	yes	0.831	0.832	0.800	0.813	1.039	0.927	0.202	0.000
	ProtCOVER-W	no	0.802	0.812	0.703	0.807	0.965	0.941	0.240	0.000
	ProtCOVER-W	yes	0.837	0.841	0.788	0.812	1.046	0.935	0.171	0.000
	(i)+(iii) gate	no	0.796	0.805	0.697	0.800	0.952	0.944	0.250	0.002
	(i)+(iii) gate	yes	0.838	0.845	0.809	0.797	1.051	0.947	0.168	0.005
	ProtCOVER-W + gate	no	0.803	0.813	0.705	0.808	0.970	0.936	0.238	0.005
	ProtCOVER-W + gate	yes	0.838	0.842	0.789	0.814	1.054	0.927	0.170	0.007
	Support-only abst. (B5)	no	0.803	0.813	0.705	0.808	0.969	0.937	0.238	0.005
	Support-only abst. (B5)	yes	0.838	0.842	0.790	0.813	1.052	0.929	0.170	0.007
	Random abst. (B5)	no	0.803	0.813	0.705	0.808	0.970	0.937	0.239	0.005
	Random abst. (B5)	yes	0.838	0.842	0.790	0.813	1.052	0.929	0.170	0.007

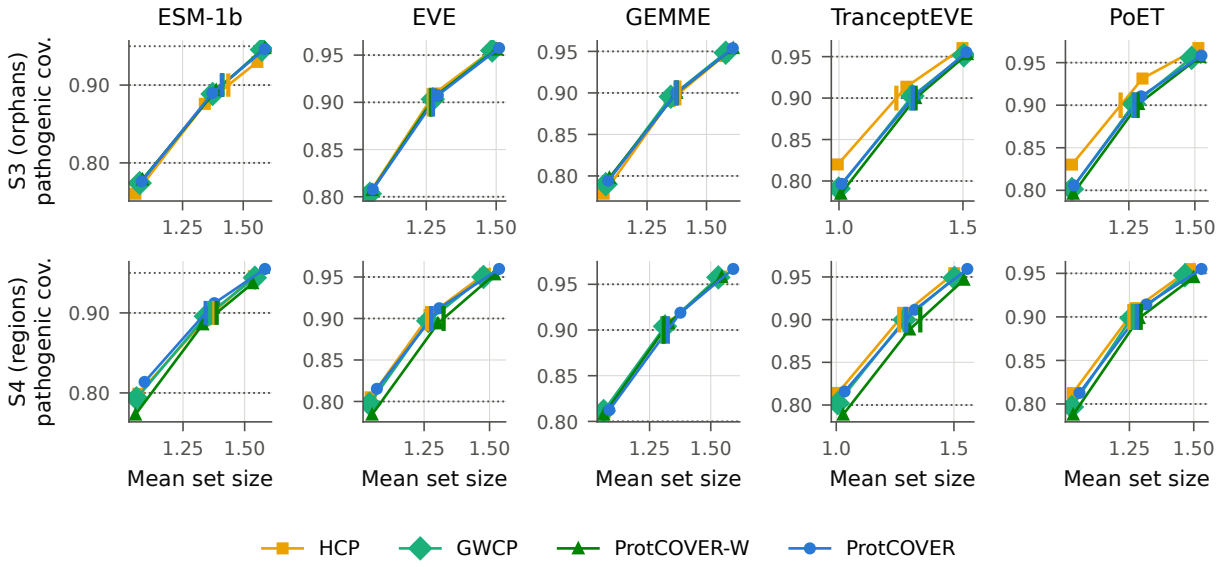


Fig. 3. Pathogenic coverage against mean set size for $\alpha \in \{0.2, 0.1, 0.05\}$ (left to right along each curve), class-conditional methods, on the two shifts defined independently of the descriptors. Dotted lines mark the three targets. A short vertical bar on the 0.90 line marks where a method's curve crosses pathogenic coverage 0.90, found by linear interpolation between its two neighboring levels: the set size the method needs for that coverage, the quantity compared in the last row of Table 2. The methods lie on one coverage-size curve, so a method that covers more also returns larger sets.

Table 3. Family-weighted pathogenic / benign coverage (Cov_1 / Cov_0) of class-conditional methods at $\alpha = 0.1$ (target 0.90), per predictor, mean over ten seeds.

Setting	Method	ESM-1b	EVE	GEMME	TranceptEVE	PoET
S1 (exchangeable)	Mondrian split (B2)	0.749 / 0.899	0.778 / 0.898	0.755 / 0.895	0.928 / 0.837	0.790 / 0.877
	HCP (B3)	0.903 / 0.903	0.904 / 0.901	0.898 / 0.898	0.900 / 0.901	0.899 / 0.903
	GWCP (B4)	0.902 / 0.901	0.902 / 0.899	0.897 / 0.897	0.900 / 0.900	0.899 / 0.901
	ProtCOVER-W	0.897 / 0.904	0.904 / 0.902	0.895 / 0.899	0.902 / 0.902	0.897 / 0.906
	ProtCOVER-D	0.906 / 0.906	0.901 / 0.908	0.901 / 0.903	0.908 / 0.907	0.904 / 0.906
	ProtCOVER	0.903 / 0.905	0.903 / 0.904	0.901 / 0.900	0.906 / 0.905	0.901 / 0.907
S2, $\beta = 1$	Mondrian split (B2)	0.655 / 0.929	0.709 / 0.915	0.712 / 0.906	0.920 / 0.839	0.795 / 0.856
	HCP (B3)	0.838 / 0.927	0.867 / 0.916	0.879 / 0.909	0.884 / 0.894	0.908 / 0.875
	GWCP (B4)	0.899 / 0.904	0.888 / 0.909	0.891 / 0.904	0.895 / 0.894	0.891 / 0.897
	ProtCOVER-W	0.910 / 0.905	0.895 / 0.909	0.902 / 0.908	0.897 / 0.898	0.897 / 0.905
	ProtCOVER-D	0.908 / 0.910	0.898 / 0.914	0.899 / 0.911	0.901 / 0.907	0.903 / 0.904
	ProtCOVER	0.909 / 0.911	0.897 / 0.915	0.903 / 0.912	0.899 / 0.906	0.899 / 0.908
S2, $\beta = 2$	Mondrian split (B2)	0.619 / 0.941	0.686 / 0.917	0.700 / 0.910	0.921 / 0.839	0.806 / 0.840
	HCP (B3)	0.798 / 0.939	0.855 / 0.917	0.869 / 0.914	0.880 / 0.891	0.920 / 0.853
	GWCP (B4)	0.888 / 0.905	0.877 / 0.906	0.887 / 0.897	0.888 / 0.889	0.895 / 0.873
	ProtCOVER-W	0.902 / 0.915	0.896 / 0.916	0.897 / 0.915	0.907 / 0.899	0.909 / 0.891
	ProtCOVER-D	0.911 / 0.919	0.897 / 0.910	0.890 / 0.918	0.894 / 0.907	0.918 / 0.892
	ProtCOVER	0.929 / 0.918	0.924 / 0.927	0.921 / 0.925	0.931 / 0.922	0.920 / 0.926
S3 (orphan families)	Mondrian split (B2)	0.678 / 0.915	0.772 / 0.895	0.724 / 0.903	0.943 / 0.823	0.817 / 0.854
	HCP (B3)	0.876 / 0.908	0.909 / 0.899	0.890 / 0.905	0.914 / 0.892	0.931 / 0.879
	GWCP (B4)	0.889 / 0.905	0.903 / 0.898	0.895 / 0.900	0.903 / 0.913	0.901 / 0.900
	ProtCOVER-W	0.894 / 0.908	0.907 / 0.901	0.900 / 0.908	0.900 / 0.921	0.901 / 0.910
	ProtCOVER-D	0.894 / 0.909	0.906 / 0.899	0.907 / 0.908	0.905 / 0.916	0.912 / 0.903
	ProtCOVER	0.890 / 0.910	0.907 / 0.902	0.900 / 0.909	0.902 / 0.917	0.910 / 0.905
S4 (held-out regions)	Mondrian split (B2)	0.732 / 0.904	0.780 / 0.893	0.763 / 0.897	0.938 / 0.820	0.798 / 0.869
	HCP (B3)	0.891 / 0.899	0.904 / 0.900	0.907 / 0.898	0.908 / 0.887	0.910 / 0.895
	GWCP (B4)	0.896 / 0.902	0.897 / 0.902	0.904 / 0.895	0.900 / 0.895	0.899 / 0.901
	ProtCOVER-W	0.886 / 0.905	0.894 / 0.913	0.907 / 0.891	0.888 / 0.913	0.899 / 0.918
	ProtCOVER-D	0.885 / 0.916	0.909 / 0.909	0.916 / 0.901	0.890 / 0.908	0.909 / 0.920
	ProtCOVER	0.912 / 0.905	0.912 / 0.903	0.919 / 0.905	0.912 / 0.904	0.914 / 0.915

Table 4. Ablations: pathogenic coverage Cov_1 at $\alpha = 0.1$, mean (minimum) over the five predictors, class-conditional versions.

Configuration	S2, $\beta = 1$	S2, $\beta = 2$	S3	S4
Components				
(i) family-level calibration (HCP)	0.875 (0.838)	0.864 (0.798)	0.904 (0.876)	0.904 (0.891)
(i)+(ii), conditional on $\hat{w}$ (ProtCOVER)	0.901 (0.897)	0.925 (0.920)	0.902 (0.890)	0.914 (0.912)
(i)+(ii), conditional on descriptors (ProtCOVER-D)	0.901 (0.898)	0.902 (0.890)	0.905 (0.894)	0.902 (0.885)
(i)+(ii), reweighting by $\hat{w}$ (ProtCOVER-W)	0.900 (0.895)	0.902 (0.896)	0.900 (0.894)	0.895 (0.886)
(i)+(iii) gate	0.875 (0.838)	0.864 (0.798)	0.904 (0.876)	0.904 (0.891)
ProtCOVER-W + (iii) gate	0.900 (0.895)	0.904 (0.898)	0.901 (0.894)	0.895 (0.886)
(ii) without (i): variant-level weighting	0.789 (0.740)	0.787 (0.723)	0.807 (0.740)	0.798 (0.737)
Descriptors for $\hat{w}$ (ProtCOVER-W)				
sequence similarity only	0.871 (0.850)	0.853 (0.799)	0.899 (0.892)	0.906 (0.903)
alignment depth only	0.899 (0.892)	0.901 (0.872)	0.899 (0.878)	0.903 (0.893)
embedding distance only	0.879 (0.845)	0.866 (0.806)	0.904 (0.872)	0.895 (0.882)
the three support groups	0.900 (0.894)	0.904 (0.898)	0.900 (0.894)	0.896 (0.887)
main set + predictor score summaries	0.903 (0.894)	0.898 (0.889)	0.902 (0.895)	0.894 (0.887)
Weight clipping c (default 20; ProtCOVER-W)				
$c = 5$	0.899 (0.894)	0.889 (0.870)	0.900 (0.892)	0.904 (0.897)
$c = 100$	0.900 (0.895)	0.908 (0.904)	0.901 (0.894)	0.883 (0.874)
no clipping	0.900 (0.895)	0.908 (0.904)	0.901 (0.894)	0.837 (0.779)

Table 5. Pooling, feature classes and the share of calibration families that fit $\hat{w}$, over the 25 shifted cells at $\alpha = 0.1$ (class-conditional versions). Shortfall: worse-class shortfall, mean (maximum) over cells. Set size: mean. s.d. Cov_1: seed-to-seed standard deviation of pathogenic coverage, mean over cells. Abstention: largest fraction of test families with an infinite threshold in any cell. Margin: cells where ProtCOVER's shortfall is smaller / larger than the variant's by more than 0.01. Paired: cells where ProtCOVER's worse-class coverage is higher / lower than the variant's with a paired 95% interval that excludes zero. Variants were declared in the configuration before they were run.

Variant	Shortfall	Set size	s.d. Cov_1	Abstention	Margin	Paired
ProtCOVER, features $(1, \hat{w})$, ratio fitted on a third (default)	0.001 (0.010)	1.357	0.018	0.3%	-	-
one variant per family and class (Gibbs et al.)	0.002 (0.016)	1.355	0.020	0.3%	1/0	7/1
features $(1, \hat{w}, d)$	0.000 (0.003)	1.380	0.017	0.3%	0/0	0/18
ratio fitted on 20% of calibration families	0.002 (0.015)	1.349	0.019	1.1%	0/0	13/1
ratio fitted on 50%	0.001 (0.008)	1.375	0.019	1.5%	0/0	3/10
ProtCOVER-D, features $(1, d)$ (default)	0.003 (0.015)	1.324	0.017	0.0%	3/0	-
features $(1, d, d^2)$	0.001 (0.007)	1.338	0.016	0.4%	0/0	6/8
ProtCOVER-W, ratio fitted on a third (default)	0.003 (0.014)	1.335	0.023	0.0%	2/0	-
ratio fitted on 20%	0.005 (0.019)	1.331	0.024	0.0%	3/0	-
ratio fitted on 50%	0.003 (0.020)	1.349	0.025	0.0%	3/0	-

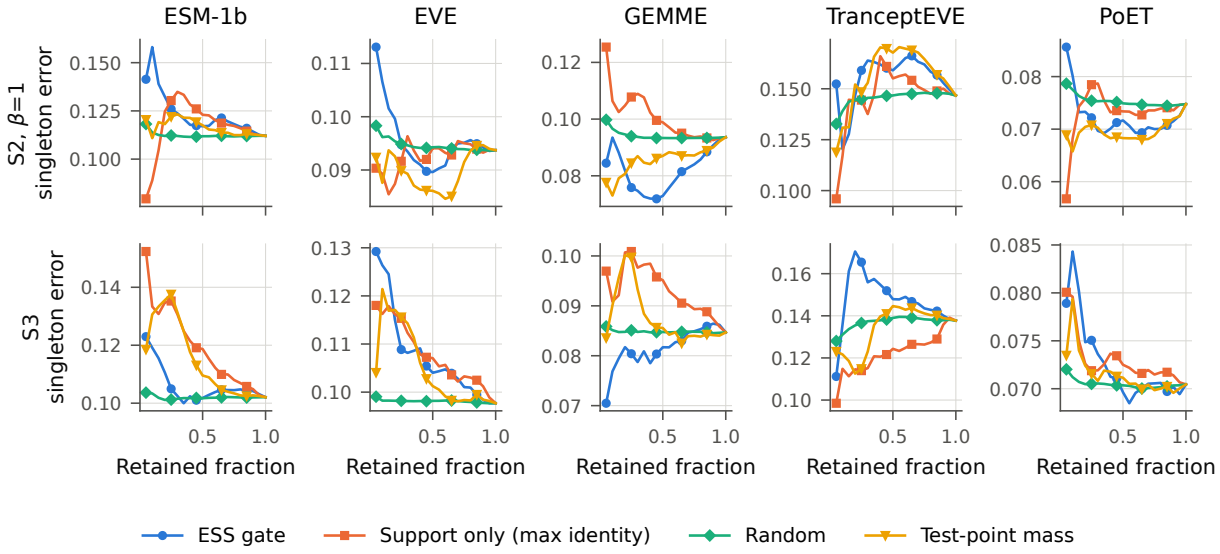


Fig. 4. Risk-coverage of abstention rules applied to ProtCOVER-W's class-conditional sets ($\alpha = 0.1$, mean over seeds): error of retained singleton sets as whole families are abstained in order of the score. Random abstention is averaged over 100 random family orders. The gate's decision depends only on label-free quantities, so it abstains on the same families whichever variant of (ii) it wraps.

Table 6. Paired differences in pathogenic coverage Cov_1: ProtCOVER minus each baseline.

		Cov_1 (pathogenic)		
Setting	Predictor	− HCP	− GWCP	− ProtCOVER-W
S1	ESM-1b	+0.001 [−0.003, +0.005]	+0.002 [−0.002, +0.006]	+0.006 [+0.003, +0.011]
	EVE	−0.001 [−0.004, +0.001]	+0.001 [−0.002, +0.003]	−0.001 [−0.004, +0.001]
	GEMME	+0.003 [−0.001, +0.006]	+0.004 [+0.001, +0.007]	+0.005 [+0.002, +0.009]
	TranceptEVE	+0.005 [+0.003, +0.008]	+0.006 [+0.003, +0.009]	+0.004 [+0.001, +0.007]
	PoET	+0.002 [−0.001, +0.005]	+0.002 [−0.001, +0.005]	+0.004 [+0.001, +0.007]
S2, $\beta = 1$	ESM-1b	+0.071 [+0.065, +0.078]	+0.010 [+0.004, +0.017]	−0.001 [−0.008, +0.005]
	EVE	+0.031 [+0.027, +0.036]	+0.009 [+0.005, +0.013]	+0.002 [−0.002, +0.007]
	GEMME	+0.023 [+0.020, +0.027]	+0.012 [+0.008, +0.016]	+0.001 [−0.002, +0.005]
	TranceptEVE	+0.015 [+0.012, +0.019]	+0.005 [+0.002, +0.008]	+0.002 [−0.000, +0.005]
	PoET	−0.008 [−0.011, −0.006]	+0.008 [+0.005, +0.010]	+0.003 [+0.000, +0.005]
S2, $\beta = 2$	ESM-1b	+0.131 [+0.124, +0.139]	+0.042 [+0.035, +0.049]	+0.027 [+0.021, +0.034]
	EVE	+0.069 [+0.064, +0.075]	+0.047 [+0.042, +0.053]	+0.028 [+0.024, +0.032]
	GEMME	+0.052 [+0.047, +0.057]	+0.034 [+0.029, +0.039]	+0.025 [+0.020, +0.029]
	TranceptEVE	+0.052 [+0.048, +0.056]	+0.043 [+0.039, +0.048]	+0.024 [+0.021, +0.028]
	PoET	−0.000 [−0.003, +0.003]	+0.025 [+0.021, +0.029]	+0.010 [+0.007, +0.013]
S3	ESM-1b	+0.014 [+0.011, +0.017]	+0.001 [−0.002, +0.004]	−0.004 [−0.006, −0.001]
	EVE	−0.002 [−0.004, +0.000]	+0.004 [+0.002, +0.006]	+0.000 [−0.001, +0.002]
	GEMME	+0.010 [+0.007, +0.012]	+0.005 [+0.002, +0.007]	+0.000 [−0.002, +0.002]
	TranceptEVE	−0.012 [−0.014, −0.009]	−0.001 [−0.003, +0.002]	+0.002 [+0.000, +0.004]
	PoET	−0.021 [−0.024, −0.018]	+0.009 [+0.007, +0.012]	+0.009 [+0.007, +0.012]
S4	ESM-1b	+0.021 [+0.018, +0.025]	+0.016 [+0.013, +0.020]	+0.026 [+0.022, +0.031]
	EVE	+0.008 [+0.006, +0.011]	+0.015 [+0.012, +0.019]	+0.018 [+0.015, +0.022]
	GEMME	+0.012 [+0.009, +0.014]	+0.015 [+0.012, +0.018]	+0.012 [+0.009, +0.014]
	TranceptEVE	+0.004 [+0.002, +0.006]	+0.012 [+0.009, +0.014]	+0.023 [+0.020, +0.026]
	PoET	+0.004 [+0.002, +0.007]	+0.016 [+0.013, +0.018]	+0.016 [+0.013, +0.019]

Table 7. Paired differences in benign coverage Cov_0: ProtCOVER minus each baseline.

		Cov_0 (benign)		
Setting	Predictor	− HCP	− GWCP	− ProtCOVER-W
S1	ESM-1b	+0.002 [−0.000, +0.004]	+0.004 [+0.002, +0.006]	+0.001 [−0.001, +0.003]
	EVE	+0.003 [+0.002, +0.005]	+0.005 [+0.004, +0.007]	+0.003 [+0.001, +0.004]
	GEMME	+0.001 [−0.000, +0.003]	+0.003 [+0.001, +0.004]	+0.001 [−0.000, +0.002]
	TranceptEVE	+0.004 [+0.002, +0.007]	+0.005 [+0.003, +0.008]	+0.003 [+0.001, +0.006]
	PoET	+0.004 [+0.001, +0.007]	+0.006 [+0.004, +0.009]	+0.001 [−0.002, +0.004]
S2, $\beta = 1$	ESM-1b	−0.016 [−0.019, −0.015]	+0.006 [+0.004, +0.009]	+0.006 [+0.004, +0.008]
	EVE	−0.001 [−0.003, −0.000]	+0.006 [+0.005, +0.008]	+0.005 [+0.004, +0.007]
	GEMME	+0.002 [+0.001, +0.004]	+0.008 [+0.006, +0.010]	+0.004 [+0.002, +0.005]
	TranceptEVE	+0.012 [+0.010, +0.014]	+0.011 [+0.010, +0.013]	+0.007 [+0.006, +0.009]
	PoET	+0.033 [+0.029, +0.036]	+0.010 [+0.007, +0.013]	+0.002 [−0.001, +0.006]
S2, $\beta = 2$	ESM-1b	−0.020 [−0.022, −0.018]	+0.014 [+0.012, +0.016]	+0.004 [+0.001, +0.006]
	EVE	+0.011 [+0.009, +0.013]	+0.021 [+0.019, +0.023]	+0.012 [+0.010, +0.014]
	GEMME	+0.011 [+0.009, +0.013]	+0.028 [+0.026, +0.031]	+0.010 [+0.008, +0.012]
	TranceptEVE	+0.030 [+0.027, +0.034]	+0.032 [+0.029, +0.036]	+0.022 [+0.019, +0.025]
	PoET	+0.073 [+0.069, +0.078]	+0.053 [+0.049, +0.058]	+0.035 [+0.031, +0.039]
S3	ESM-1b	+0.002 [+0.000, +0.003]	+0.005 [+0.004, +0.006]	+0.002 [+0.001, +0.003]
	EVE	+0.004 [+0.003, +0.005]	+0.004 [+0.003, +0.005]	+0.001 [+0.000, +0.002]
	GEMME	+0.004 [+0.003, +0.005]	+0.009 [+0.007, +0.010]	+0.001 [+0.000, +0.002]
	TranceptEVE	+0.025 [+0.023, +0.028]	+0.004 [+0.002, +0.007]	−0.004 [−0.005, −0.002]
	PoET	+0.026 [+0.023, +0.029]	+0.004 [+0.002, +0.006]	−0.005 [−0.007, −0.003]
S4	ESM-1b	+0.006 [+0.004, +0.008]	+0.003 [+0.001, +0.006]	−0.000 [−0.002, +0.002]
	EVE	+0.003 [+0.001, +0.004]	+0.001 [−0.001, +0.002]	−0.010 [−0.012, −0.008]
	GEMME	+0.008 [+0.006, +0.009]	+0.010 [+0.008, +0.012]	+0.014 [+0.012, +0.017]
	TranceptEVE	+0.016 [+0.014, +0.019]	+0.009 [+0.006, +0.011]	−0.010 [−0.012, −0.007]
	PoET	+0.020 [+0.018, +0.023]	+0.014 [+0.011, +0.017]	−0.003 [−0.005, +0.000]

Table 8. Paired differences in mean set size (variant-weighted): ProtCOVER minus each baseline.

		Mean set size		
Setting	Predictor	− HCP	− GWCP	− ProtCOVER-W
S1	ESM-1b	−0.013 [−0.019, −0.006]	−0.009 [−0.015, −0.002]	−0.003 [−0.008, +0.004]
	EVE	+0.010 [+0.006, +0.015]	+0.016 [+0.012, +0.021]	+0.006 [+0.003, +0.010]
	GEMME	−0.004 [−0.008, −0.000]	−0.001 [−0.005, +0.003]	−0.001 [−0.005, +0.003]
	TranceptEVE	+0.010 [−0.000, +0.019]	+0.013 [+0.003, +0.022]	+0.003 [−0.007, +0.012]
	PoET	+0.009 [+0.002, +0.015]	+0.013 [+0.006, +0.019]	+0.009 [+0.002, +0.016]
S2, $\beta = 1$	ESM-1b	+0.079 [+0.069, +0.091]	−0.020 [−0.031, −0.010]	−0.051 [−0.061, −0.040]
	EVE	+0.041 [+0.035, +0.047]	+0.031 [+0.025, +0.037]	+0.021 [+0.016, +0.027]
	GEMME	+0.047 [+0.041, +0.053]	+0.030 [+0.025, +0.037]	−0.003 [−0.008, +0.002]
	TranceptEVE	+0.052 [+0.046, +0.059]	+0.034 [+0.027, +0.042]	+0.022 [+0.016, +0.029]
	PoET	+0.020 [+0.014, +0.026]	+0.026 [+0.020, +0.033]	+0.000 [−0.005, +0.007]
S2, $\beta = 2$	ESM-1b	+0.188 [+0.173, +0.204]	+0.055 [+0.042, +0.069]	+0.012 [+0.001, +0.024]
	EVE	+0.170 [+0.158, +0.184]	+0.159 [+0.148, +0.173]	+0.114 [+0.105, +0.126]
	GEMME	+0.150 [+0.138, +0.163]	+0.123 [+0.112, +0.135]	+0.080 [+0.073, +0.090]
	TranceptEVE	+0.220 [+0.206, +0.237]	+0.206 [+0.192, +0.223]	+0.154 [+0.142, +0.168]
	PoET	+0.149 [+0.137, +0.163]	+0.174 [+0.161, +0.189]	+0.117 [+0.106, +0.129]
S3	ESM-1b	+0.031 [+0.027, +0.037]	−0.001 [−0.005, +0.005]	−0.015 [−0.020, −0.010]
	EVE	+0.012 [+0.010, +0.013]	+0.018 [+0.016, +0.020]	+0.006 [+0.005, +0.007]
	GEMME	+0.029 [+0.026, +0.032]	+0.024 [+0.020, +0.027]	−0.001 [−0.004, +0.002]
	TranceptEVE	+0.031 [+0.027, +0.035]	+0.007 [+0.003, +0.011]	−0.006 [−0.009, −0.003]
	PoET	−0.005 [−0.009, −0.002]	+0.027 [+0.025, +0.030]	+0.010 [+0.007, +0.013]
S4	ESM-1b	+0.040 [+0.033, +0.047]	+0.035 [+0.027, +0.042]	+0.047 [+0.039, +0.055]
	EVE	+0.037 [+0.029, +0.047]	+0.042 [+0.034, +0.051]	+0.005 [−0.001, +0.012]
	GEMME	+0.052 [+0.043, +0.064]	+0.063 [+0.053, +0.074]	+0.048 [+0.039, +0.058]
	TranceptEVE	+0.043 [+0.036, +0.050]	+0.044 [+0.036, +0.052]	+0.022 [+0.013, +0.031]
	PoET	+0.041 [+0.034, +0.048]	+0.058 [+0.052, +0.065]	+0.028 [+0.023, +0.035]

Table 9. Paired differences in singleton error: ProtCOVER minus each baseline.

		Singleton error		
Setting	Predictor	− HCP	− GWCP	− ProtCOVER-W
S1	ESM-1b	−0.002 [−0.005, +0.001]	−0.003 [−0.006, −0.001]	−0.001 [−0.003, +0.002]
	EVE	+0.002 [+0.001, +0.005]	+0.001 [−0.001, +0.003]	+0.003 [+0.001, +0.005]
	GEMME	+0.000 [−0.002, +0.002]	−0.001 [−0.003, +0.001]	+0.001 [−0.001, +0.002]
	TranceptEVE	+0.001 [−0.009, +0.012]	+0.000 [−0.009, +0.011]	+0.002 [−0.008, +0.013]
	PoET	+0.008 [+0.005, +0.011]	+0.007 [+0.004, +0.010]	+0.008 [+0.006, +0.011]
S2, $\beta = 1$	ESM-1b	+0.007 [+0.004, +0.010]	−0.010 [−0.014, −0.007]	−0.011 [−0.015, −0.007]
	EVE	−0.000 [−0.002, +0.001]	−0.001 [−0.002, +0.001]	+0.001 [−0.001, +0.002]
	GEMME	+0.002 [+0.001, +0.004]	−0.002 [−0.003, −0.001]	−0.000 [−0.002, +0.001]
	TranceptEVE	+0.001 [−0.002, +0.004]	+0.008 [+0.004, +0.012]	+0.010 [+0.006, +0.014]
	PoET	−0.013 [−0.015, −0.011]	−0.004 [−0.006, −0.002]	+0.000 [−0.002, +0.002]
S2, $\beta = 2$	ESM-1b	+0.010 [+0.006, +0.015]	−0.014 [−0.018, −0.010]	−0.005 [−0.008, −0.001]
	EVE	−0.004 [−0.006, −0.001]	−0.007 [−0.009, −0.004]	+0.001 [−0.001, +0.004]
	GEMME	+0.002 [−0.001, +0.005]	−0.014 [−0.017, −0.010]	+0.000 [−0.003, +0.003]
	TranceptEVE	+0.004 [−0.003, +0.010]	+0.008 [+0.000, +0.015]	+0.018 [+0.011, +0.024]
	PoET	−0.026 [−0.029, −0.022]	−0.018 [−0.021, −0.015]	−0.007 [−0.010, −0.005]
S3	ESM-1b	−0.002 [−0.003, −0.001]	−0.005 [−0.007, −0.004]	−0.004 [−0.005, −0.002]
	EVE	−0.003 [−0.004, −0.002]	−0.004 [−0.005, −0.003]	−0.001 [−0.002, −0.000]
	GEMME	−0.002 [−0.004, −0.001]	−0.004 [−0.006, −0.003]	−0.001 [−0.003, +0.000]
	TranceptEVE	+0.004 [+0.001, +0.008]	−0.001 [−0.003, +0.002]	−0.001 [−0.003, +0.000]
	PoET	−0.004 [−0.006, −0.003]	−0.001 [−0.002, +0.001]	+0.003 [+0.001, +0.004]
S4	ESM-1b	−0.002 [−0.004, +0.000]	−0.001 [−0.003, +0.001]	−0.001 [−0.003, +0.002]
	EVE	+0.001 [−0.001, +0.002]	−0.001 [−0.002, +0.000]	+0.001 [−0.000, +0.003]
	GEMME	+0.002 [+0.001, +0.003]	−0.000 [−0.002, +0.001]	−0.002 [−0.004, −0.000]
	TranceptEVE	−0.004 [−0.011, +0.001]	−0.013 [−0.021, −0.006]	−0.024 [−0.032, −0.016]
	PoET	−0.005 [−0.006, −0.003]	−0.004 [−0.006, −0.003]	+0.001 [−0.001, +0.003]